

Week 10 – Certificates and Cryptographic Methods

Date	20.11.2026
Learning outcomes	LO.2, 4
Duration	3 hours
Prerequisites	AES-GCM, HMAC, key derivation and what TLS does, from Week 3 ; being able to run the <code>openssl</code> command in a terminal
Labs	code/week-10 – 2 demos; requires OpenSSL 3; <code>sh demo.sh</code> in each demo folder



This week's working demo

`code/week-10/01-pki-zincir` – root->intermediate->server chain with OpenSSL; `verify` errors without the intermediate certificate, OK once it's added. · `code/week-10/02-imza-dogrulama` – Ed25519 signature/verification: the correct signature is accepted, a tampered one is rejected; the `==1` trap.

To run: `sh demo.sh` (Linux/WSL) or build with CMake and run from `bin/`. Entirely synthetic and safe; it does not harm the student's computer.



Run the demo yourself – step by step (copy-paste)

These two demos use **OpenSSL** and a shell script; on Windows open **WSL** or **Git Bash**. From the `code` folder:

```
# WSL / Linux / Git Bash
cd week-10/01-pki-zincir
sh demo.sh           # kök CA → ara CA → sunucu sertifikası üretir ve zinciri doğrular

cd ../02-imza-dogrulama
sh demo.sh           # Ed25519 ile imzala → doğrula → tek bayt kurcala → doğrulama reddetsin
```

Expected output: The first demo builds a three-link **chain**; `openssl verify` finds the chain **valid**, and **verification fails once the intermediate certificate is removed** (this week's "missing intermediate certificate" rule). The second demo signs a file and verifies it (**OK**), then changes a single byte, and verification **rejects** it – this is how signature integrity catches tampering.

**By the end of this week you will be able to**

1. Choose an **algorithm**, **key length**, and **mode** for an application and base the choice on current standards (NIST SP 800-57, SP 800-131A).
2. Compare block cipher **modes** (ECB, CBC, CTR, GCM) and **padding**; explain why the padding oracle attack is possible.
3. Use **HMAC** correctly; combine encryption and the MAC in the right order and protect against **replay**.
4. Perform encryption, signing, and key exchange with **RSA** (OAEP, PSS) and **elliptic-curve** (Ed25519, X25519) algorithms; show why signed Diffie–Hellman is necessary.
5. Explain **PKI** components (CA, RA, chain, trust store); build and validate a **root** → **intermediate** → **leaf** certificate chain with OpenSSL.
6. Explain certificate **revocation** (CRL, OCSP, OCSP stapling) and know the rationale for storing keys in a software HSM (**SoftHSM**, **PKCS#11**).

**Class schedule (timing plan for the instructor)**

Time	Section	What happens
0:00–0:15	1	Map of cryptography; algorithm and key-length selection
0:15–0:50	2–3	Block cipher modes and padding; MAC, HMAC, encrypt-then-MAC, replay
0:50–1:00	Break	
1:00–1:30	4–5	RSA (OAEP, PSS) and elliptic curves; creating and verifying digital signatures (OpenSSL)
1:30–1:50	6	Key exchange: Diffie–Hellman, man-in-the-middle, signed DH
1:50–2:00	Break	
2:00–2:40	7–9	PKI components; X.509 structure; building a certificate chain with OpenSSL – class activity
2:40–2:55	10–11	Revocation (CRL, OCSP); key storage (PKCS#11 , SoftHSM, HSM)
2:55–3:00	12+	Look at post-quantum cryptography, project step, self-check

Prepare the lab in advance

This week's commands require **OpenSSL 3** (the `openssl` that ships with Git for Windows, or your package manager on WSL/Linux). Produce all files in an empty working folder; the keys you generate are **for the course only** and must never be used on any real system. The SoftHSM section is optional.

```
openssl version
mkdir C:\Courses\h10; cd C:\Courses\h10

sudo apt install -y openssl softhsm2 opensc
openssl version
mkdir -p ~/h10 && cd ~/h10
```

Ethics rule — applies every week in this course

All of this week's keys and certificates are ones we generate ourselves, **synthetic**, and for the lab only. Adding a root certificate you generated to the trust store of someone else's computer, or producing a fake certificate carrying the name of a real certificate authority, is unacceptable. Man-in-the-middle examples are only ever done on your own computer, on `localhost`.

0. Basic concepts (from scratch)

This section prepares you for the week. It first briefly recalls the earlier topics this week builds on; it then defines each of this week's concepts in one sentence and links it to the section where it is explained in full.

What we bring from earlier weeks

- **Symmetric and asymmetric encryption** — symmetric encryption runs fast with a single key shared by both parties; asymmetric encryption uses a public/private key pair, which makes key distribution easy but is slow ([Week 3, §2.2](#)). This week we go deeper into the asymmetric side's mathematics (RSA, ECC — §4) and the symmetric side's correct choice of mode and padding (§2).
- **AEAD** — authenticated encryption gives confidentiality and integrity together, in a single call, by producing a tag that also proves the data has not changed ([Week 3, §2.1](#)). This week we show why the mode inside AEAD (GCM) and the alternatives you should stay away from (ECB, CBC) give such different security (§2).
- **Digest (hash), MAC, and digital signature** — a digest is a one-way fingerprint computed from data; a MAC proves a message has not changed using a shared key, and a digital signature does the same with a public/private key pair ([Week 3, §2.4](#)). This week we cover HMAC's internal construction and the correct order for combining it with encryption (§3), and the practical pitfalls of signature verification (§5).
- **Forward secrecy** — deleting a session's derived key right after use, so that even if today's key leaks, past sessions cannot be decrypted ([Week 3, §7.1](#)). This week we build the Diffie–Hellman key exchange and its authentication — the mechanism underneath forward secrecy — from scratch (§6).
- **TLS certificate validation** — the client checking a server certificate's chain, validity dates, intended usage, and name ([Week 3, §9.2](#)). This week we build the PKI, X.509, and revocation (CRL/OCSP) machinery this validation rests on, from the ground up (§7–10).

This week's concept map

Concept	In one sentence	Detail
Block cipher modes and padding	A block cipher encrypts a fixed-size block; the way we apply it to a longer message is called a mode, and if the message isn't a multiple of the block size it is completed with padding; the choice of mode and padding matters as much as the algorithm itself.	§2
MAC and HMAC	A MAC is a function that produces a short tag from a secret key and a message, proving the message hasn't changed; HMAC is the standard way to build one from a hash function using two nested calls.	§3
Asymmetric cryptography: RSA and ECC	RSA relies on the difficulty of factoring large integers, while elliptic curve cryptography (ECC) relies on the difficulty of the discrete logarithm problem on a curve; ECC gives the same security with a much shorter key.	§4
Digital signature	A mechanism proving a message was created by the holder of a specific private key and has not been altered, verifiable by anyone with the public key; unlike a MAC, it also provides non-repudiation.	§5
Diffie–Hellman (DH)	A key-exchange method that lets two parties agree on a shared secret over an insecure channel without ever sharing a prior secret; without authentication it is open to a man-in-the-middle attack.	§6
PKI and CA	Public key infrastructure (PKI) is the set of rules, roles, and documents that lets you verify, through certificate authorities (CAs), that a public key really belongs to the party who claims it.	§7
Certificate and X.509	X.509 is the standard format for certificates that bind a public key to an identity and carry a CA's signature; it contains the subject, public key, validity dates, and extensions (SAN, key usage).	§8
CRL and OCSP	A CRL is a CA-signed list of revoked certificates; OCSP is a protocol for querying a certificate's revocation status on the spot.	§10
HSM, PKCS#11, SoftHSM	An HSM is a tamper-resistant hardware module that generates and operates on keys without ever releasing them; PKCS#11 is the standard interface for talking to such modules, and SoftHSM is a software simulation of one for testing.	§11
Post-quantum cryptography (PQC)	New algorithms built on different mathematical problems, such as lattices, as a hedge against the assumption that a sufficiently large quantum computer could efficiently solve the problems RSA and elliptic curves rely on.	§12

How do the concepts connect?

The terms above are not a random list; each one sits on top of the previous ones. The table below shows the relationship "to understand this, I first need to know that" — if you forget a term, go back to the "prerequisite" column in the table.

Term	Prerequisite	Covered in depth in
Block cipher / mode	Symmetric cryptography	Section 2
Padding	Block cipher / mode	Section 2
AEAD	Block cipher, MAC	Sections 2–3
MAC / HMAC	Digest (hash), symmetric key	Section 3
Digital signature	Asymmetric cryptography, digest	Sections 4–5
RSA / ECC	Asymmetric cryptography	Section 4
Diffie–Hellman	Asymmetric cryptography, signature	Section 6
PKI / CA	Digital signature, trust	Section 7
Certificate / X.509	PKI, signature, digest	Sections 8–9
CRL / OCSP	Certificate, CA	Section 10
HSM / PKCS#11	The concept of a private key	Section 11
PQC	The math problems RSA/ECC rely on	Section 12

The direction in which this table is read matters: **to understand the term at the start of a row, you must first know the term in the "prerequisite" column.** For example, before reading the "certificate revocation" (CRL/OCSP) section, the concepts of "certificate" and "CA" need to already be settled; otherwise the question "why does the CA sign the revocation list?" is left hanging.



How should you read this week?

Sections build on each other **in order**: 1 → 2 → 3 (symmetric tools), 4 → 5 → 6 (asymmetric tools), 7 → 8 → 9 → 10 (fitting these into PKI), 11 (storing the key), 12 (a look ahead). If you skip a section and move on, sentences in a later section like "recall X from Week 3" or "Y from the previous section" will be meaningless. Re-read a section that feels hard a second time, then move on.

1. The map of cryptography and algorithm selection

We ended Week 9 with a warning: obfuscating code does **not** store a key, it only makes the surroundings harder to work with ([Week 9, §3](#)). This week we move on to the actual fix — instead of hiding keys, we protect them **for real** with the right algorithm choice, the right protocol, and, where needed, hardware-backed storage (PKI, HSM). In Week 3 we used cryptography as a **tool** to protect data in transit, at rest, and in use ([Week 3, §1](#)); this week we look inside those same tools: which algorithm does which job, why some were abandoned, how to choose, and how a certificate authority (CA) fits into this structure. The book's Chapters 4–8 and 10–11 cover these topics in more than 55 recipes; since most of the algorithms have changed since 2003, we give each one its current equivalent.

Simetrik ve asimetrik: hangisi ne için?

pratikte HİBRİT: asimetrikle anahtar kur, simetrikle veri şifrele

SİMETRİK (AES)

tek anahtar
ÇOK HIZLI

sorun: anahtarı nasıl paylaşırsın?

ASİMETRİK (RSA/EC)

açık + gizli anahtar çifti
YAVAŞ

anahtar dağıtımı ve İMZA çözülür

TLS tam olarak bunu yapar: ECDHE ile anahtar, AES-GCM ile veri.

A brief history: keys, certificates, and PKI

- **1976–77** — Diffie–Hellman and **RSA**: public-key cryptography solves the problem of "talking securely with someone you've never met."
- **1988** — the **X.509** certificate format is standardized; identity is carried by a **CA signature**.
- **1995** — the first commercial CAs (VeriSign) and **PKI** become widespread; the revocation mechanisms **CRL** and **OCSP** follow.
- **2014** — **Heartbleed** and POODLE tighten the TLS ecosystem; **2015 Let's Encrypt** makes HTTPS widespread with free certificates; **2018** TLS 1.3.
- **2022–2024** — NIST selects the **post-quantum** algorithms (Kyber/ML-KEM, Dilithium/ML-DSA).

All of this week's rules (the right mode, the right padding, chain validation, revocation) come from the painful lessons of this timeline.

Five basic jobs, five tool families

Job	Tool	Current choice	Abandoned
Confidentiality (symmetric)	Block and stream ciphers	AES-256-GCM, ChaCha20-Poly1305	DES, 3DES, RC4, Blowfish
Integrity + identity (symmetric)	MAC	HMAC-SHA-256, CMAC, GCM's tag	CBC-MAC (variable-length), MD5-based MACs
Fingerprint	Hash function	SHA-256, SHA-384, SHA-3	MD5, SHA-1
Key exchange	Diffie–Hellman, RSA key transport	X25519, ECDHE (P-256)	1024-bit DH, RSA key transport (removed in TLS 1.3)
Signature (asymmetric)	Digital signature	Ed25519, ECDSA P-256, RSA-PSS ≥ 3072	DSA, RSA PKCS#1 v1.5 signature (in new designs)

Key length: security level

The key lengths of different algorithms cannot be compared directly. NIST SP 800-57 compares them at an **equivalent security level** (in bits):

Security level	Symmetric	Digest (collision resistance)	RSA / DH	Elliptic curve
112 bit	3DES (legacy systems only)	SHA-224	2048	224
128 bit	AES-128	SHA-256	3072	256 (P-256, Ed25519/X25519)
192 bit	AES-192	SHA-384	7680	384
256 bit	AES-256	SHA-512	15360	512

Today, the floor for a new design is the **128-bit security level**. This means 3072 bits for RSA, 256 bits for elliptic curve; 2048-bit RSA is not considered sufficient for new use after 2030 in NIST's transition timeline. The table's most important lesson is that elliptic curves provide the same security with **much shorter** keys; that is why they are preferred on mobile and embedded systems.

Its counterpart in the book

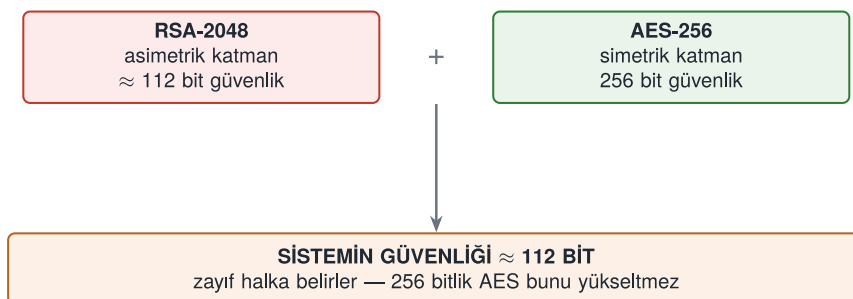
The book covers algorithm and key-length selection in Recipes 5.2 and 5.3, digest and MAC selection in 6.3 and 6.4, and public-key algorithm and length selection in 7.2 and 7.3. The principle it recommends still holds today: **choose a well-studied, standard algorithm and use it through a high-level, error-resistant API** (Recipe 5.16). Only the algorithms themselves have changed.

The weakest-link rule

A system uses more than one cryptographic layer: key exchange with one algorithm, data encryption with another. The system's security level equals the **lowest** of these layers — not the strongest. If you transport a key with RSA-2048 and encrypt the data with AES-256, the attacker doesn't try to break 256-bit AES; they target the ≈ 112 -bit RSA layer. So evaluate the layers **together**, by the lowest level, not one at a time.

En zayıf halka kuralı

sistemin gücü, en güçlü parçası kadar değildir



Katmanları ayrı ayrı değil, en düşük güvenlik seviyesine göre değerlendirin.

One rule: don't write your own crypto — or your own protocol

Even if every algorithm is correct, combining them (choosing what order to encrypt and MAC in, where to get the nonce from, how to separate the keys) requires a separate kind of expertise. All of this week's examples will show that almost all real-world mistakes are **in the combination, not the algorithm**. Wherever possible, use ready-made, proven constructions (TLS 1.3, AEAD, HPKE, Noise).

Worked example: how big is the difference between "128 bit" and "112 bit"?

The bit counts in the table can stay abstract; let's make them concrete. A security level of "n bits" means the attacker needs, on average, 2^{n-1} attempted operations (2^n is used as the upper bound). The difference between RSA-2048 (112 bit) and AES-128 (128 bit) is far larger than it looks:

Hand calculation (exponential growth)

```
2^112 ≈ 5,19 × 10^33    (RSA-2048's equivalent difficulty)
2^128 ≈ 3,40 × 10^38    (AES-128's key space)
2^128 / 2^112 = 2^16 = 65.536 → breaking AES-128 is 65.536 TIMES harder than breaking RSA-2048 at
"equivalent" difficulty
```

Let's convert this into time. Let's take, as an **optimistic** upper bound, 10^{18} (quintillion; exascale supercomputer) operations per second for today's fastest supercomputers — in real attacks this number is much lower, because breaking a cipher is not a single arithmetic operation but a far more costly one. Still, let's calculate with the upper bound:

How long does 2^{112} operations take at 10^{18} operations/second?

```
2^112 operations ÷ 10^18 operations/second ≈ 1,6 × 10^8 years  (= 165 million years)
Comparison: recorded human history ≈ 5.000 years → 165 million years is ~33.000 times that
2^128 operations ÷ 10^18 operations/second ≈ 1,08 × 10^13 years
Comparison: the age of the universe ≈ 1,38 × 10^10 years → 2^128 takes ~780 times the age of the
universe
```

The lesson to draw has two parts. First, **even 112 bits cannot practically be brute-forced today**; the reason RSA-2048 is considered "insufficient" for use after 2030 is not brute force, but (a) hardware getting cheaper over time, (b) the need to protect data that must be kept for a long time with a **margin** against future attacks, and © the fact that quantum computers will be able to break RSA far faster than brute force (see Section 12). Second, the seemingly small difference between 112 bits and 128 bits (16 bits) cuts the attacker's work down to **one in 65,536** — you must read bit counts **exponentially**, not linearly.

⚡ Common mistake: thinking it's enough to say 'we use AES'

Writing only the algorithm's name (AES) in an algorithm inventory says nothing. AES-128-ECB and AES-256-GCM are members of the same family but worlds apart in terms of security: one (ECB) leaks patterns (Section 2), the other (GCM) also provides integrity. In an algorithm inventory, **algorithm + mode + key length + the standard it rests on** must be written together (the book's Recipe 5.16 and this week's "S8" template in Section 13).

✓ Rule

Document a cryptographic choice not just by the algorithm's name, but by the **(algorithm, key length, mode, library, standard)** quintuple. "We use AES" is not an answer; "AES-256-GCM, OpenSSL EVP, NIST SP 800-38D" is.

In [Week 3](#) we only saw the "which tool do I use when" part of this map (AEAD, the hybrid scheme, random numbers). This week we see **why** the same map is drawn this way – which mathematical problem gives which security, why each algorithm was abandoned. The next two sections look inside the symmetric tools (mode, padding, MAC); Sections 4–6 look inside the asymmetric tools (RSA, ECC, DH); Sections 7–10 will look at how these come together in PKI.

2. Block cipher modes and padding (Recipe 5.4, 5.11)

AES is a **block cipher**: it turns exactly a 16-byte block into another 16-byte block. To encrypt a longer message you need a **mode of operation** that chains the blocks together. The choice of mode is as important as the choice of algorithm; recall [the ECB penguin from Week 3](#).

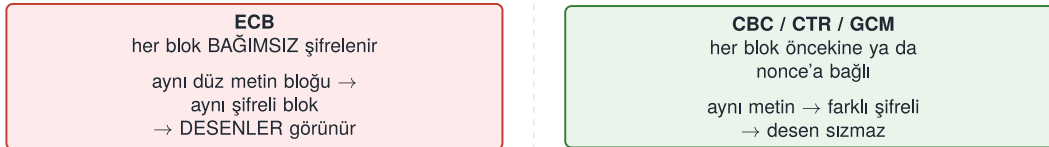
Nonce tekrarı neden felaket?

gizlilik VE bütünlük birlikte çöker



ECB neden asla kullanılmaz?

ünlü "ECB pengueni" bu yüzden görünür kalır



Kip seçimi şifrelemenin kendisi kadar önemlidir; varsayılan tercih AEAD (GCM).

Mode	How it works	Confidentiality	Integrity	Parallel	Padding	Assessment
ECB	Each block is encrypted independently	X (leaks patterns)	X	✓	Required	Do not use
CBC	Each block is XORed with the previous ciphertext block	✓ (if the IV is unpredictable)	X	Decryption only	Required	With a separate MAC; watch for the padding oracle
CTR	A counter is encrypted to produce a keystream	✓ (if the nonce is unique)	X	✓	Not required	With a separate MAC
GCM	CTR + multiplication-based authentication	✓	✓	✓	Not required	Recommended (AEAD)
ChaCha20-Poly1305	Stream cipher + Poly1305 MAC	✓	✓	✓	Not required	Recommended fast on devices without AES hardware support

Padding and the padding oracle

ECB and CBC require the message length to be a multiple of the block size; the missing part is filled in with **padding**. The most common padding is **PKCS#7**: if the number of missing bytes is n , n bytes each with the value n are appended (e.g., if 3 bytes are missing, `03 03 03`). The decrypting party reads the last byte and drops that many bytes; but first it checks whether the padding is **valid**.

This is where the problem begins: if a decryption failure is **distinguishable** as "invalid padding" versus "invalid MAC" (a different error message, a different response time), an attacker can change bytes of the ciphertext and, by observing the server's reaction, decrypt the message byte by byte. This is called the **padding oracle** attack; it was described in 2002, and it kept reappearing in the following years in web frameworks and in TLS (POODLE, Lucky Thirteen).

Defences:

1. **Use AEAD** (GCM, ChaCha20-Poly1305): there is no padding; if the tag doesn't match, nothing is decrypted.
2. If CBC is mandatory, **encrypt first, then MAC** (next section) and verify the MAC **before checking the padding**.
3. Return all decryption errors as a **single, identical** error, in the same amount of time.



How it's done in the field

In the algorithm inventory of a certified mobile payment library, AES is seen used in CBC and CTR modes with a separate SHA-256-based MAC; for the cryptogram computation of the payment schemes, a DES-based MAC algorithm resting on ISO/IEC 9797-1 (with the padding methods the schemes specify) is used. These choices come from scheme specifications that must be complied with. For internal communication that the scheme does not specify, AEAD is preferred today ([the critical reading from Week 3](#)).

Worked example: tracing **PKCS#7** padding by hand

Let's not leave the rule "if 3 bytes are missing, `03 03 03` is appended" abstract; let's trace it step by step with a real OpenSSL run. Let's encrypt the 13-byte text `"MERHABA DUNYA"` with AES-128-CBC using a fixed 16-byte key and an IV that counts up from zero (only for this example; in a real system the IV is generated randomly for every message):

Tracing PKCS#7 padding by hand (real output)

```
K=00112233445566778899aabbccddeeff # 16 bytes (32 hex characters) – fixed only for this demonstration
IV=000102030405060708090a0b0c0d0e0f # 16 bytes – fixed only for this demonstration
printf 'MERHABA DUNYA' | openssl enc -aes-128-cbc -K "$K" -iv "$IV" -out cikti.bin
xxd -p cikti.bin
```

Output (the actually produced value):

```
4b354371f98acde5bed95d59a8135538
```

The input was 13 bytes; AES's block size is 16 bytes; the output is also exactly **16 bytes** (32 hex characters) — so a 3-byte padding was added. To see the padding's **content**, let's decrypt with `-nopad` (we would normally never do this; here only for teaching purposes):

```
openssl enc -d -aes-128-cbc -K "$K" -iv "$IV" -in cikti.bin -nopad | xxd
```

```
00000000: 4d45 5248 4142 4120 4455 4e59 4103 0303  MERHABA DUNYA...
```

The last three bytes are `03 03 03` — exactly as the **PKCS#7** rule says: as many bytes as the missing count ($16 - 13 = 3$) were added, each with that value. A normal `openssl enc -d` (without the no-padding flag) automatically reads and drops these three bytes and returns exactly `"MERHABA DUNYA"` to you — verify this too:

```
openssl enc -d -aes-128-cbc -K "$K" -iv "$IV" -in cikti.bin
```

```
MERHABA DUNYA
```

The padding oracle attack kicks in exactly at this point: the attacker changes the last bytes of the ciphertext and sends it to the server; if the server returns "invalid padding" (e.g., the last byte is `07` but the preceding 7 bytes are not `07`) and "padding valid but MAC invalid" as **different** errors, the attacker can distinguish these two cases and reconstruct the plaintext byte by byte — without ever needing the key.

Worked example: seeing why ECB leaks patterns

In ECB, every 16-byte block is encrypted **independently**; the same plaintext block always produces the same ciphertext block. Let's encrypt a plaintext that repeats over 48 bytes ("AAAAAAAAAAAAAAAA" three times) with AES-128-ECB:

Repeating block in ECB = repeating ciphertext (real output)

```
K2=$(openssl rand -hex 16)
printf 'AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA' | head -c 48 > duz.txt # 48 bytes = 3
blocks, all the same
openssl enc -aes-128-ecb -K "$K2" -in duz.txt -out cikti_ecb.bin -nopad
xxd cikti_ecb.bin
```

The output of an example run (since the key is random, you will see different byte values, but the **pattern will stay the same**):

```
00000000: f5cf 4879 d8fe 9c90 1975 2de6 eb9f 59a1 ..Hy.....u-...Y.
00000010: f5cf 4879 d8fe 9c90 1975 2de6 eb9f 59a1 ..Hy.....u-...Y.
00000020: f5cf 4879 d8fe 9c90 1975 2de6 eb9f 59a1 ..Hy.....u-...Y.
```

All three blocks are **identical**. Even an observer who doesn't know the key learns the information "these three blocks contain the same plaintext" — which is often a leak in its own right (e.g., an image's background, a form's fixed fields, a repeating record structure). If you repeat the same experiment with AES-128-CBC (with a random IV), all three blocks come out **different**; because each block is chained by being XORed with the previous one.

Common mistake: saying 'we used AES, so we're secure'

A call made in a codebase without specifying the mode (e.g., `Cipher.getInstance("AES")` in Java) can **silently fall back to ECB** depending on the library (in Java, the "AES" shorthand means "AES/ECB/PKCS5Padding"). This is a mistake that is easy to miss in code review but leaks real data in production. Rule: the mode must **always be written explicitly**, and calls using ECB must be rejected at the static-analysis/build stage.

Rule

Never leave the mode name to the default; write it out with its full name, like `AES-256-GCM`. Never use ECB under any circumstances (file encryption, a DB field, even for "unimportant" data) — it always leaks the pattern. If CBC is mandatory, protect the padding with the encrypt-then-MAC rule from Section 3.

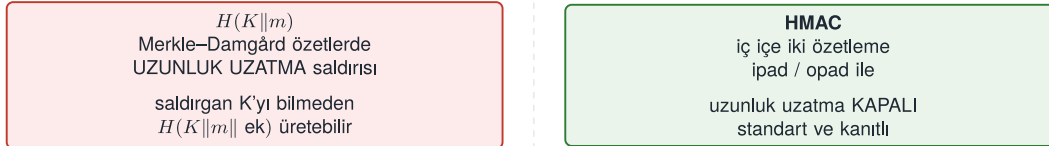
In [Week 2's](#) threat modelling, "integrity" was an asset-protection goal; this section made concrete **why** that goal cannot be achieved by encryption alone. The next section covers the tool that correctly provides integrity — MAC/HMAC — and the right order for combining it.

3. MAC, HMAC, and combining integrity with encryption (Recipe 6.4, 6.10, 6.18, 6.21)

How does HMAC work?

Neden HMAC, neden $H(K||m)$ değil?

kendi MAC'inizi kurmayın



Doğrulamayı sabit zamanlı karşılaştırmayla yapın (CRYPTO_memcmp).

Encrypt-then-MAC: doğru sıra

alıcı geçersiz metni hiç çözmez → dolgu kâhini kapanır



Yanlış sıra (önce çöz, sonra doğrula) dolgu hatası sızdırır ve oracle açar.

Encrypt-then-MAC: because the receiver never decrypts invalid ciphertext, the "error leak" that a padding oracle needs never arises. AES-GCM does these two steps in a single operation.

A **MAC** (message authentication code) produces a short tag from a secret key and a message; someone who doesn't know the key cannot produce a valid tag. **HMAC** is the standard way of building a MAC from a hash function (SHA-256) (RFC 2104):

$$\text{HMAC}(K, m) = H((K \oplus \text{opad}) \parallel H((K \oplus \text{ipad}) \parallel m))$$

The two nested digests remove the **length-extension** weakness of the plain $H(K \parallel m)$ construction: in Merkle–Damgård-structured digests like SHA-256, if $H(K \parallel m)$ is known, $H(K \parallel m \parallel \text{extra})$ can be computed without knowing the key. So **don't derive your own MAC from a digest**; use HMAC (the warning in Recipe 6.19).

HMAC-SHA-256 with OpenSSL

```
printf 'tutar=100;alici=TR00' | openssl dgst -sha256 -mac HMAC -macopt hexkey:$(openssl rand -hex 32)
```

Three ways to combine encryption with a MAC

Method	How?	Example	Assessment
Encrypt-then-MAC	Encrypt first, then take the MAC of the ciphertext	IPsec ESP, TLS's EtM extension	The correct way: the MAC is verified first; if invalid, the ciphertext is never decrypted
MAC-then-Encrypt	Take the MAC of the plaintext, encrypt both together	Old TLS (CBC modes)	Exposed to the padding oracle; avoid
Encrypt-and-MAC	Encrypt the plaintext, separately attach the MAC of the plaintext	Old SSH	The MAC can leak information about the plaintext; avoid

AEAD modes make this choice correctly on your behalf. If you must do your own combination (Recipe 6.18): use **two separate keys** (one for encryption, one for the MAC; both can be derived from a single secret with HKDF), follow the encrypt-then-MAC order, include the IV and context information in the MAC as well, and make the comparison constant-time.

Replay: a valid but old message

A MAC proves that a message has **not been changed**, but not that it is **new**. An attacker can record a valid "send 100 TL" message and resend it ten times; the MAC is valid every time. The defences the book describes in Recipe 6.21:

Method	How?	Note
Counter	Every message carries a number larger than the previous counter; the receiver stores the last counter	Simplest and most reliable; the counter must be inside the MAC
Timestamp	The message must fall within a short validity window	Requires clock synchronization; needs an extra measure against repeats within the window
Nonce (one-time value)	The server gives a new value for every request, the client includes it in the MAC	The server remembers the values used



The counter in payments

In contactless card transactions, every transaction is signed with a cryptogram that contains the card's **transaction counter** (ATC); the server does not accept the same counter a second time. In mobile payment libraries, each one-time key is also used in only one transaction, and the counter is kept in a database. This is why the "rollback of the usage counter" threat has the asset-protection class **I+** in [Week 2's](#) threat table.

Worked example: computing HMAC by hand and seeing the avalanche effect

Without leaving the rules abstract, let's trace a real HMAC-SHA-256 computation end to end. Let's use a fixed (only for this example) 32-byte key and compute the HMAC of a payment instruction:

HMAC-SHA-256 (real output)

```
ANAHTAR=0000000000000000000000000000000000000000000000000000000000000001 # 32 bytes, fixed only for  
this demonstration  
printf 'tutar=100;alici=TR00' | openssl dgst -sha256 -mac HMAC -macopt hexkey:$ANAHTAR
```

```
SHA2-256(stdin)= a08fb1159871e3b8110dedf3e3524cab7bea06f4795f2119394e11c42ac5f9a1
```

Now let the attacker change the amount from 100 to 900; with the real key, the original message's HMAC changes as follows:

```
printf 'tutar=900;alici=TR00' | openssl dgst -sha256 -mac HMAC -macopt hexkey:$SANAHTAR
```

```
SHA2-256(stdin)= 8a786eacbef7863d44533b9ebcbcf288e5136204080258e838e283f1f91a46be
```

A single-character change (`1` $\rightarrow$ `9`) produced **completely different** 32 bytes for the tag — this is a good hash function's "avalanche effect": a single-bit change in the input changes roughly half of the output. Since the attacker doesn't know the key, they cannot change the message and produce a **valid** new tag; the receiver compares the tags and catches the mismatch.

Constant-time comparison: why `memcmp` is not enough

On the receiving side, HMAC verification requires **comparing** the computed tag with the tag that arrived. An ordinary `memcmp(a, b, 32)` compares bytes left to right and **stops at the first differing byte**; this means the function's running time varies depending on how many bytes matched. An attacker who can make enough attempts over a network can guess the correct tag **byte by byte** from microsecond differences in response times (a timing attack).

Incorrect and correct comparison

```
/* WRONG: an early exit creates a timing difference */
if (memcmp(hesaplanan, gelen, 32) == 0) { /* accept */ }

/* CORRECT: constant-time, always walks through all 32 bytes */
if (CRYPTO_memcmp(hesaplanan, gelen, 32) == 0) { /* accept */ } /* OpenSSL */
/* or: sodium_memcmp (libsodium), hmac.compare_digest (Python) */
```

⚡ Common mistake: comparing a MAC/tag with `==` or `memcmp`

If a comparison like `strcmp(hesaplanan_imza, gelen_imza) == 0` is seen in a web service's API signature verification code, this is not a theoretical but a **usable** timing-attack risk: with enough requests (thousands to tens of thousands), an attacker can discover the correct signature byte by byte. Rule: **no** secret-value comparison is ever done with an early-exit function.

 Rule

Always compare secret values such as a MAC, a signature, or a password hash with a constant-time function (`CRYPTO_memcmp`, `sodium_memcmp`, the language's own "constant-time compare" function). Don't write your own comparison function; this is another appearance of this week's "don't write your own crypto" rule (Section 1).

In [Week 3](#) we introduced AEAD as "the tool that gives confidentiality + integrity in a single package"; this section showed the correct order and the correct comparison for when you are forced to provide integrity **separately** (an old protocol where you can't use AEAD, a hardware constraint, etc.). The next section moves on to the asymmetric tools that give what a MAC cannot — **non-repudiation**.

4. Asymmetric cryptography: RSA and elliptic curves (Recipe 7.1–7.3)

Symmetric cryptography's only difficulty is that the two parties must share the same key **in advance**. In asymmetric (public-key) cryptography, each party has a **key pair**: a **public key** given to everyone, and a **private key** known only to its owner. What is encrypted with the public key can only be decrypted with the private key; what is signed with the private key can be verified by everyone with the public key.

RSA dolgusu: hangisi nerede?

dolgu, algoritmanın güvenliğinin ayrılmaz parçasıdır

OAEP RSA ile ŞİFRELEME rastgeleleştirilmiş dolgu ham RSA asla kullanılmaz	PSS RSA ile İMZA rastgeleleştirilmiş dolgu PKCS#1 v1.5 yerine tercih
---	--

Eliptik eğride karşılıkları: X25519 (anahtar değişimi), Ed25519 (imza).

Asymmetric operations are hundreds, thousands of times slower than symmetric ones. This is why asymmetric cryptography is used for only two jobs (Recipe 7.1): **carrying or agreeing on a short symmetric key** and **signing**. The data itself is always protected with a symmetric cipher ([the hybrid scheme from Week 3](#)).

RSA: with the right padding

RSA's mathematics in its plain form ("textbook RSA") is insecure: the same message always gives the same ciphertext, and its mathematical structure is open to attacks. Secure use depends on the **correct padding scheme**:

Job	Correct scheme	Obsolete scheme	Why?
Encryption	RSA-OAEP (SHA-256)	PKCS#1 v1.5 encryption	Padding-oracle attacks against v1.5 encryption (since 1998, with recurring variants)
Signature	RSA-PSS	PKCS#1 v1.5 signature	v1.5 signing is still widespread and not broken, but PSS is the scheme with a security proof

The book's "raw" RSA operations in Recipes 7.10–7.13 are there to help you understand how unpadded RSA works; they are not used in production. The warning in Recipe 7.14 still matters, though: combining **sign-and-encrypt** needs care; if the signature is not bound to the recipient and context of the encrypted content, the recipient can re-encrypt the signed message and forward it to someone else.

RSA-3072 with OpenSSL 3: key, OAEP encryption, PSS signature


```
openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:3072 -out rsa_ozel.pem
openssl pkey -in rsa_ozel.pem -pubout -out rsa_acik.pem

# Encrypt a short session key with OAEP (SHA-256)
openssl rand -out oturum.key 32
openssl pkeyutl -encrypt -pubin -inkey rsa_acik.pem -in oturum.key -out oturum.enc \
    -pkeyopt rsa_padding_mode:oaep -pkeyopt rsa_oaep_md:sha256

# Sign and verify with PSS
openssl dgst -sha256 -sign rsa_ozel.pem -sigopt rsa_padding_mode:pss -out belge.sig belge.txt
openssl dgst -sha256 -verify rsa_acik.pem -sigopt rsa_padding_mode:pss -signature belge.sig belge.txt
```

Elliptic curves: the same security, a shorter key

Elliptic-curve cryptography (ECC) provides the security level of 3072-bit RSA with a 256-bit key; its operations are faster, and its keys and signatures are much shorter.

Algorithm	Job	Note
Ed25519	Signature	Deterministic (does not require a random number), fast, carefully designed against side channels
X25519	Key agreement	The default for protocols like TLS 1.3, SSH, Signal
ECDSA (P-256)	Signature	Widespread and standard; but requires a unique and secret random value on every signature
ECDH (P-256)	Key agreement	Widespread on FIPS-approved systems



ECDSA and the random number

ECDSA uses a secret random value (k) on every signature. If the same k is used in two different signatures, or if it can be predicted, the **private key** can be computed from the two signatures. In 2010, a game console's signing key was exposed exactly this way because the manufacturer used the same k on every signature; in 2013, some mobile crypto wallets suffered the same kind of loss because of a weak random generator. This is the most striking proof of [Week 3's](#) sentence "random numbers are the invisible foundation of cryptography." Ed25519 and the deterministic ECDSA in RFC 6979 remove this risk.

Ed25519 signature with OpenSSL 3

```
openssl genpkey -algorithm ED25519 -out ed_ozel.pem
openssl pkey -in ed_ozel.pem -pubout -out ed_acik.pem
openssl pkeyutl -sign -inkey ed_ozel.pem -rawin -in belge.txt -out belge.ed.sig
openssl pkeyutl -verify -pubin -inkey ed_acik.pem -rawin -in belge.txt -sigfile belge.ed.sig
```

Worked example: measuring RSA-3072 against Ed25519

Let's confirm the sentence "ECC gives the same security with a smaller key" with real bytes. Both are at ~128-bit security level (see the table in Section 1):

Measuring key and signature sizes (real output)

```

wc -c rsa_acik.pem ed_acik.pem
openssl dgst -sha256 -sign rsa_ozel.pem -sigopt rsa_padding_mode:pss -out belge.rsa.sig belge.txt
openssl pkeyutl -sign -inkey ed_ozel.pem -rawin -in belge.txt -out belge.ed.sig
wc -c belge.rsa.sig belge.ed.sig

```

```

636 rsa_acik.pem
116 ed_acik.pem
384 belge.rsa.sig
64 belge.ed.sig

```

The public key file (PEM, Base64-encoded) is **636 bytes** for RSA, **116 bytes** for Ed25519 — roughly 5.5 times smaller. The signature is **384 bytes** for RSA (3072 bits = 384 bytes; an RSA signature is exactly the size of the modulus), **64 bytes** for Ed25519 — 6 times smaller. When you consider the data an IoT device sends on every handshake, the number of certificates an embedded system stores in flash, or the number of signatures carried by every transaction on a blockchain, this difference accumulates and grows; this is why ECC is the default in mobile/embedded systems and in new protocols (TLS 1.3, SSH, Signal).

Worked example: why is OAEP's randomness necessary?

In Section 2 we saw that ECB produces the same ciphertext for the same plaintext block, and that this is a pattern leak. "Textbook RSA" (raw, unpadded) carries the same disease: the same message always produces the same ciphertext. OAEP fixes this by mixing a **random** value into the message before encryption. Let's encrypt the same message twice with OAEP:

OAEP: same message, two different ciphertexts (real output)

```

echo MERHABA > kisa.txt
openssl pkeyutl -encrypt -pubin -inkey rsa_acik.pem -in kisa.txt -out c1.bin \
  -pkeyopt rsa_padding_mode:oaep -pkeyopt rsa_oaep_md:sha256
openssl pkeyutl -encrypt -pubin -inkey rsa_acik.pem -in kisa.txt -out c2.bin \
  -pkeyopt rsa_padding_mode:oaep -pkeyopt rsa_oaep_md:sha256
cmp c1.bin c2.bin && echo "AYNI" || echo "FARKLI"

```

```

c1.bin c2.bin differ: char 1, line 1
FARKLI

```

Same public key, same plaintext, but **different** ciphertext — because OAEP uses a new random value on every encryption (the RSA counterpart of Section 1's "random numbers are the invisible foundation of cryptography" principle). If you run the same experiment with unpadded ("raw") RSA, the two outputs come out **byte-for-byte identical**; mathematically, $c = m^e \bmod n$ is a deterministic operation. This is why raw RSA is never used directly for encryption.

**Common mistake: using an RSA key for both encryption and signing**

Using the same RSA key pair for both "encrypt the data" and "sign the data" overloads two different mathematical operations onto the same key; in some attack scenarios (e.g., a signing request being turned into a decryption request by appearing to sign a carefully chosen "plaintext") this sharing can be abused. Rule: **a key's usage purpose is singular** — the certificate's `keyUsage` field (Section 8) enforces this too: `keyEncipherment` and `digitalSignature` are separate bits, and a single key should not carry both.

✓ Rule

Always use **OAEP** for encryption and **PSS** for signing; **PKCS#1** v1.5 is only considered when compatibility with an old system is mandatory, and even then only for signing (never for new encryption designs). In a new design, prefer Ed25519/X25519 over RSA where possible.

In Week 3 we used asymmetric cryptography (RSA or ECDHE back then) as a black box in TLS's key setup. This section opened up that black box: which padding, which curve, which size. The next section goes deeper into the **signature** leg of these tools; then in Section 6 we move on to the **key exchange** leg.

5. Digital signature: creation, verification, and where it's used

A **digital signature** proves that a message (1) was created by the owner of a specific private key, and (2) has not been altered afterward. Unlike a MAC, verification is done with the **public key**: anyone can verify, but only the owner of the private key can sign. This is why a signature provides **non-repudiation** (the R in STRIDE), a MAC does not.

Dijital imza: oluşturma ve doğrulama

en sık hata: `if(r)` ile denetlemek



API 1=başarı, 0=başarısız, <0=hata döner; `if(r)` hatayı da "doğru" sayar.

Where are signatures used?

Use	What is signed?	Where it appears in this course
Software update	The update package	Week 1's "Vault" example, threat T6
Code signing	Executable file, mobile app package	Week 6 signature verification
Certificate	Public key + identity information (the CA signs)	PKI, this week
TLS handshake	Handshake digest (the server signs)	Week 3's <code>CertificateVerify</code>
Document and transaction	Contract, payment approval	Electronic-signature regulation

Verifying a signature in code: OpenSSL EVP

Verifying an Ed25519 signature (OpenSSL 3)

```
#include <openssl/evp.h>
#include <openssl/pem.h>

/* 1: valid, 0: invalid, -1: error. Called BEFORE running the update package. */
int imza_dogrula(EVP_PKEY *acik, const unsigned char *veri, size_t n,
                const unsigned char *imza, size_t imza_n)
{
    EVP_MD_CTX *ctx = EVP_MD_CTX_new();
    int sonuc = -1;
    if (ctx == NULL) return -1;
    if (EVP_DigestVerifyInit(ctx, NULL, NULL, NULL, acik) == 1) /* Ed25519: digest is NULL */
        sonuc = EVP_DigestVerify(ctx, imza, imza_n, veri, n); /* 1 or 0 */
    EVP_MD_CTX_free(ctx);
    return sonuc == 1 ? 1 : (sonuc == 0 ? 0 : -1);
}
```

⚠ The three traps of signature verification

- The return value:** `EVP_DigestVerify` returns 1 on success, 0 if the signature is invalid, and a negative value on error. A check like `if (sonuc)` treats **a negative error value as "valid."** The check must always be `== 1`.
- What is signed:** If the signature only covers the package's content, an attacker can restore an **old and vulnerable** but validly signed package (a rollback). The version number and the target product must also be part of the signed content, and the client must reject an older version.
- Which key:** Verifying the signature with a public key that comes **bundled inside the package itself** proves nothing. The verification key must be embedded in the application beforehand (with its own integrity protected) or come from a trusted chain.

Worked example: tracing signature verification in three cases from the command line

You can see the logic of the C code above with `openssl pkeyutl -verify` too, without needing to compile – the CLI produces the same three outcomes (valid / invalid / error):

1) Correct document, correct signature

```
openssl pkeyutl -verify -pubin -inkey ed_acik.pem -rawin -in belge.txt -sigfile belge.ed.sig
echo "cikis kodu: $?"
```

```
Signature Verified Successfully
cikis kodu: 0
```

2) Document changed, signature unchanged

```
echo "MERHABA DEGISTI" > belge2.txt
openssl pkeyutl -verify -pubin -inkey ed_acik.pem -rawin -in belge2.txt -sigfile belge.ed.sig
echo "cikis kodu: $?"
```

```
Signature Verification Failure
cikis kodu: 1
```

The moment a single byte of the document changed, verification **failed** and the exit code went from 0 to 1 – if your update mechanism doesn't check this code (or if the C code interprets `EVP_DigestVerify`'s return with `if (sonuc)`), you have made a tampered package **executable**.

Rollback: the signature is valid but the content is old

Let's make the "scope of the signature" trap concrete. Suppose an application update server signs each version separately:

Version	Content	Signature
v1.0	(contains a known vulnerability)	<code>imza_v1</code> – still mathematically valid
v2.0	closes the vulnerability	<code>imza_v2</code> – valid

After the server publishes v2.0, v1.0's signature is **not revoked** – signatures do not have a revocation mechanism like CRL/OCSP (the mechanism we will see for certificates in Section 10). If an attacker who has obtained the v1.0 package presents it along with `imza_v1` to the client, the client only asks "is the signature valid?", gets the answer **yes**, and installs the vulnerable version. This is why the signed data must not be only the file's content, but the **version number + target product name**, and the client must additionally enforce, **on its own**, the rule "do not accept a version older than the highest one I have seen" – the signature does not do this for you.

**Common mistake: only asking 'is it signed or not'**

A client asking only a binary question like `is_signed(package)` (without asking which key, for which version, for which target it was signed) leaves the door open to rollback and **target confusion** attacks (a package signed for one product being presented for another).

 Rule

Signed data should always carry its **context**: version, target product/device, timestamp, or validity window. A client must ask not just "is this signature valid," but "is this signature valid **for this** context and for the **most current** version."

In [Week 1's "Vault" example](#) we saw that accepting an update package without a signature leads to the T6 threat; this section showed that even the **existence** of a signature alone is not enough — its scope must also be designed correctly. The next section covers how we use signatures and asymmetric cryptography in **key exchange** — and what goes wrong if we don't.

6. Key exchange: Diffie–Hellman and man-in-the-middle (Recipe 8.16–8.18)

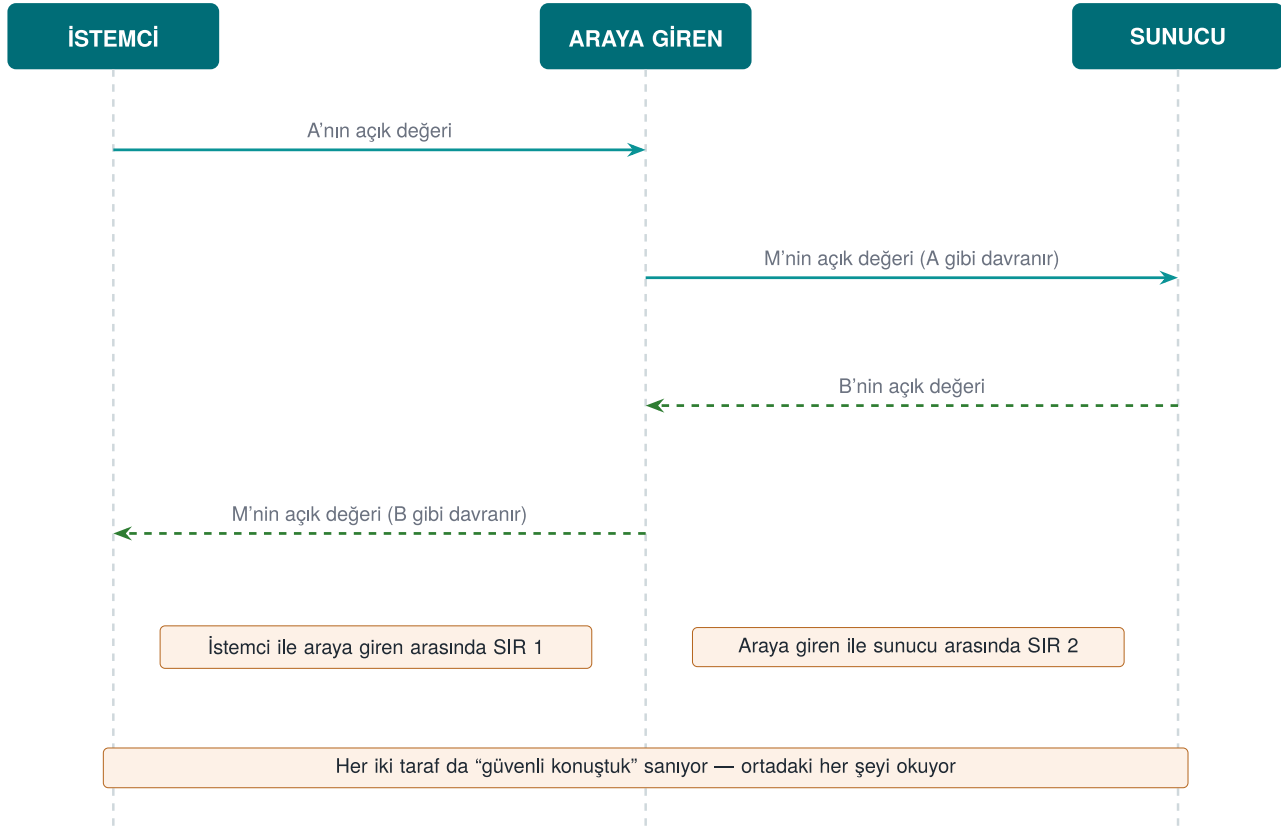
Diffie–Hellman (DH) lets two parties agree on a shared secret over an insecure channel, with no previously shared secret at all. Each party picks a private value and sends the public value derived from it; both parties compute the **same** shared secret from the other party's public value and their own private value. An eavesdropper sees the public values but cannot compute the shared secret. Today's form of this is **ECDH** over an elliptic curve (X25519).

The weakness of unauthenticated DH: man-in-the-middle

DH does not say **who** you agreed with. An attacker in the middle of the channel performs DH **separately** with each of the two parties:

Kimliksiz Diffie–Hellman: araya girme

DH kiminle anlaştığınızı SÖYLEMEZ



Çözüm: DH'yi kimlik kanıtıyla bağla (sertifika / imzalı DH) — TLS 1.3 bunu yapar.

Both parties think "I have a secure channel"; in reality every message passes through the attacker's hands.

The fix: authenticated key exchange

The public values are **signed** with a key whose identity is known in advance (the current version of Recipe 8.18's "DH and DSA together" idea: ECDHE + Ed25519/ECDSA/RSA-PSS). In TLS 1.3, the server signs the handshake digest — which also includes the ephemeral ECDHE public value — with the private key in its certificate (`CertificateVerify`); the client verifies the certificate by chain and by name ([Week 3](#)). This way:

- **Identity:** The signature proves that whoever sent the public value really is the certificate's owner.
- **Forward secrecy:** DH values are freshly generated and discarded for every session; even if the long-lived signing key is stolen later, past sessions' keys cannot be computed ([Week 3](#)).

Identity without PKI

Not every scenario has a certificate authority. The "authentication without PKI" paths the book describes in Recipe 8.19 are still used today: learning the key on first connection and warning if it later changes (TOFU, SSH), verifying the key's fingerprint over a separate channel (the "safety number" comparison in secure messaging), or password-authenticated key exchange from a secret both parties shared in advance (PAKE, Recipe 8.15; today protocols like OPAQUE, SPAKE2).

Worked example: two parties reaching the same secret with X25519

Let's confirm Diffie–Hellman's sentence "two parties reach the same shared value, without a previously shared secret" with real keys. Alice and Bob each independently generate an X25519 key pair and exchange only their **public** keys:

X25519 key agreement: Alice and Bob (real output)

```
openssl genpkey -algorithm X25519 -out alice.key
openssl genpkey -algorithm X25519 -out bob.key
openssl pkey -in alice.key -pubout -out alice.pub
openssl pkey -in bob.key -pubout -out bob.pub

# Alice: her own private key + Bob's public key
openssl pkeyutl -derive -inkey alice.key -peerkey bob.pub -out alice_sir.bin
# Bob: his own private key + Alice's public key
openssl pkeyutl -derive -inkey bob.key -peerkey alice.pub -out bob_sir.bin

xxd -p alice_sir.bin
xxd -p bob_sir.bin
cmp alice_sir.bin bob_sir.bin && echo "AYNI" || echo "FARKLI"
```

```
8c53744a000c1a6ffc2183a0c134c2cf8167110314e8a06dd3d453080bbc1376
8c53744a000c1a6ffc2183a0c134c2cf8167110314e8a06dd3d453080bbc1376
AYNI
```

The two parties, without ever knowing each other's private key, reached the **same** 32-byte shared secret, passing only public keys over the network. This shared secret is not used directly for data encryption; it is passed through a KDF (HKDF, the key-derivation topic from Week 3) and session keys are derived from it. A third-party eavesdropper sees only `alice.pub` and `bob.pub`; computing the shared secret from these (without solving the elliptic-curve discrete logarithm problem) is not possible.

Worked example: how many steps does a man-in-the-middle take to set up?

Man-in-the-middle against unauthenticated DH requires **two separate** DH agreements — the attacker's computational load is no different from the victims', just twice the operations:

1. The attacker (M) connects to Alice and sends their own public value as `pub_M1`; Alice thinks this is Bob's public value.
2. M connects to Bob and sends `pub_M2`; Bob thinks this is Alice's public value.
3. M computes `sir_A = DH(özel_M1, pub_Alice)` with Alice and `sir_B = DH(özel_M2, pub_Bob)` with Bob — obtaining **two different** shared secrets.
4. Alice sends an encrypted message with `sir_A` to "Bob" (actually M); M decrypts it, reads it, re-encrypts it with `sir_B`, and forwards it to the real Bob. Both parties believe the secret they computed is shared with the **correct** other

party.

The attacker's extra cost is only **twice the X25519 operations** (on the order of microseconds) — without authentication, this attack is essentially free computationally. This is why unsigned DH is never used alone in any modern protocol such as TLS 1.3.

Common mistake: using DH output directly as a key

Using the raw bytes produced by `pkeyutl -derive (alice_sir.bin` in the example above) directly as an AES key is a common mistake. Raw DH output is **not uniformly random** (it carries biases coming from the curve's mathematical structure) and produces a single key; whereas encryption and the MAC generally need **separate** keys (Section 3). Rule: always pass DH output through a KDF (HKDF-SHA-256) and derive as many keys as you need.

Rule

Never trust an unauthenticated DH/ECDH result; public values must always be signed or must be part of a previously validated certificate (TLS 1.3's `CertificateVerify`). Never use raw DH output directly as an encryption key; pass it through HKDF.

In [Week 3's](#) TLS 1.3 handshake we asked "what would happen if someone in the middle could swap the keys"; this section answered that question with concrete steps. The next three sections (7–9) build, from scratch, the identity infrastructure — PKI — that signed DH rests on.

7. Public key infrastructure (PKI): who trusts whom? (Recipe 10.1)

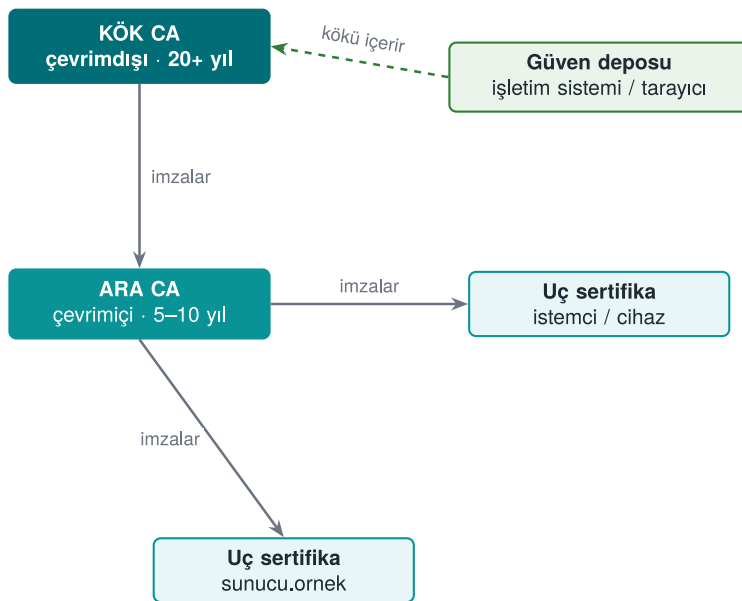
The previous section showed that public keys need to be bound to an identity through a signature. But this raises a question: how do we know that a server's public key **really** belongs to that server? **Public key infrastructure** (PKI) is the set of rules, roles, and documents that answers this question through trusted third parties.

Components

Component	Role	Example
Certificate authority (CA)	Issues a certificate by signing the public key of an identity-verified entity	Commercial and public CAs, an in-house CA
Registration authority (RA)	Verifies the identity of a certificate requester; tells the CA "issue a certificate to this person"	An organisation's HR department, an automated system that checks ownership of a domain (ACME)
Root CA	The self-signed CA at the top of the chain; usually kept offline	The roots in the operating system's and browsers' trust stores
Intermediate CA	A CA signed by the root that issues day-to-day certificates	So the root can stay offline
Leaf certificate	A certificate belonging to a server, a person, or a device	A TLS certificate for <code>sunucu.ornek</code>
Trust store	The list of trusted root certificates	The operating system's store, a store embedded in the application
Revocation services	Announce revoked certificates	CRL distribution point, OCSP responder

PKI: kim kime güvenir?

güven kökten uca akar



İstemci yalnız KÖKE güvenir; zincir kökten uca kesintisiz kurulamazsa doğrulama başarısızdır.

Why an intermediate CA? If the root CA's private key is compromised, the **entire** chain becomes untrusted, and removing the root from every trust store in the world takes years. This is why the root key sits in an HSM, offline, in a physically protected location, and is used only to sign intermediate CA certificates, in controlled meetings called a key ceremony. The intermediate CA does the day-to-day work; if it is compromised, only that intermediate CA is revoked.

Validating a certificate: four questions (Recipe 10.4)

Before a client accepts a certificate, it asks the following:

1. **Chain:** Starting from the leaf certificate, is every certificate's signature verified with the one above it's public key, and does the chain reach a root in the trust store?
2. **Validity:** Are we within the validity dates of every certificate in the chain?
3. **Usage:** Was this certificate issued for this job? (The key usage and extended key usage fields; the `CA:TRUE` basic constraint on an intermediate CA.)
4. **Name and revocation:** Is the certificate for **this** server (SAN)? Has it been revoked?

In [Week 3](#) we saw how the first and fourth questions are asked in code (`SSL_CTX_set_verify`, `SSL_set1_host`). In this section we build the same chain ourselves.

Worked example: asking the "four questions" from the command line

Let's see how a client asks the "four questions" **concretely**, using the root → intermediate → server chain we will build in Section 9 (if you don't have these three files yet, run Section 9 first; here we are only reading the same files from different angles).

Question 1 — Chain: every certificate's `issuer` field must match, **exactly**, the `subject` field of the one above it:

```
openssl x509 -in sunucu.crt -noout -issuer -subject
openssl x509 -in ara.crt -noout -issuer -subject
openssl x509 -in kok.crt -noout -issuer -subject
```

```
issuer=CN=CEN429 Lab Ara CA      subject=CN=localhost
issuer=CN=CEN429 Lab Kok CA     subject=CN=CEN429 Lab Ara CA
issuer=CN=CEN429 Lab Kok CA     subject=CN=CEN429 Lab Kok CA
```

The chain reads: `sunucu`'s issuer = `ara`'s subject; `ara`'s issuer = `kok`'s subject; `kok`'s issuer is **itself** (the root signs itself — which is why the chain stops here).

Question 2 — Validity:

```
openssl x509 -in sunucu.crt -noout -dates
openssl x509 -in sunucu.crt -noout -checkend 0
```

```
notBefore=Sep 23 06:01:46 2026 GMT
notAfter=Dec 22 06:01:46 2026 GMT
Certificate will not expire
```

`-checkend 0` asks "will this expire within 0 seconds from now?"; TLS clients perform this check automatically on every connection.

Question 3 — Usage:

```
openssl x509 -in sunucu.crt -noout -ext basicConstraints,keyUsage,extendedKeyUsage
openssl x509 -in ara.crt -noout -ext basicConstraints,keyUsage
```

```
X509v3 Basic Constraints: CA:FALSE
X509v3 Key Usage: critical / Digital Signature
X509v3 Extended Key Usage: TLS Web Server Authentication
---
X509v3 Basic Constraints: critical / CA:TRUE, pathlen:0
X509v3 Key Usage: critical / Certificate Sign, CRL Sign
```

`sunucu.crt` having `CA:FALSE` guarantees that this certificate **cannot sign another certificate** — even if an attacker obtains this certificate's private key, they cannot use it to produce and sign a new certificate. `ara.crt` is the opposite: `CA:TRUE` and `pathlen:0` (there can be no other intermediate CA below it, it can only sign leaf certificates).

Question 4 — Name: the SAN field is already visible above via `-ext subjectAltName` (`DNS:localhost`, `IP Address:127.0.0.1`); the client compares the name it connected to against this list (covered in detail in Section 8).

A client library (OpenSSL, Windows CryptoAPI, Java's `X509TrustManager`) asks these four questions **on your behalf**; disabling the `verify` functions ([Week 3's](#) `SSL_CTX_set_verify`) or swallowing errors means these four questions are never asked at all.

Common mistake: assuming 'it connected, so it must be secure'

A developer's assumption that "the TLS connection was established, so the certificate must have been validated" is wrong. The connection is also established when validation has been **turned off** ([Week 3's](#) `SSL_VERIFY_NONE` / `InsecureSkipVerify` mistake). A successful connection is not proof that the four questions were **asked**; you must **separately** check in the code that validation is turned on.

Rule

Explicitly set the validation mode (`SSL_VERIFY_PEER`, `CERT_REQUIRED`, etc.) when setting up a client library, and in a test, deliberately present an invalid certificate and verify that the connection is **rejected** — a test of "I can connect with a valid certificate" alone is not enough; you also need a test of "I cannot connect with an invalid certificate."

In the previous section we saw that signed DH proves "who I'm talking to" with a signature; this section explained how we trust **whose** signature that is (PKI, the CA hierarchy). The next two sections look at the **document itself** that carries this trust (the X.509 certificate) and at building it **by hand**.

8. The structure of an X.509 certificate

An X.509 v3 certificate consists of the **signed part** (TBSCertificate) and the CA's **signature** over that part:

X.509 sertifikasında ne var?

doğrulama bu alanların HEPSİNİ denetler

Konu (Subject) + SAN — kime ait — ad denetimi SAN'a bakar, CN'e değil

Açık anahtar (SPKI) — pinning bunu sabitler

Geçerlilik aralığı — notBefore / notAfter

Uzantılar — basicConstraints (CA:TRUE), KeyUsage, EKU

CA imzası — zincirin bir üst halkası tarafından

Yalnız “imza geçerli” demek yetmez: ad, süre, amaç ve iptal de denetlenir.

Field	Meaning
Version	v3 (required for extensions)
Serial number	Unique within the CA; revocation lists use this number
Signature algorithm	e.g., <code>ecdsa-with-SHA256</code> , <code>sha256WithRSAEncryption</code>
Issuer	The name of the CA that signed the certificate
Validity	<code>notBefore</code> – <code>notAfter</code>
Subject	The certificate's owner
Subject public key info (SPKI)	Algorithm + public key; the pinning from Week 3 uses the digest of this
Extensions	Basic constraints (<code>CA:TRUE/FALSE</code> , path length), key usage, extended key usage (server/client authentication, code signing), SAN (alternative names: DNS names, IP), CRL distribution point, OCSP address (AIA)

Certificates are stored in binary **DER** encoding, or in **PEM**, its Base64-wrapped text form (Recipe 7.16, 7.17): files starting with `-----BEGIN CERTIFICATE-----` are PEM.

⚠ Name checking is done against the SAN, not the CN

Older applications used to look for the server name in the subject's **CN** (Common Name) field. Today browsers and RFC 6125 look **only** at the **SAN** extension; a server certificate without a SAN is rejected by modern clients. Don't forget to add a SAN when generating your own certificates.

Worked example: reading a real certificate field by field

Let's read the `openssl x509 -text` output of the `sunucu.crt` we will generate in Section 9, matching it against every field in the table above (this is exactly the real output of the command `openssl x509 -in sunucu.crt -noout -text`):

openssl x509 -in sunucu.crt -noout -text (real output)

```
Certificate:
Data:
  Version: 3 (0x2)
  Serial Number:
    35:e2:f6:8d:d2:e3:90:f3:74:ff:16:a9:ec:5f:07:e9:21:69:76:e7
  Signature Algorithm: ecdsa-with-SHA256
  Issuer: CN=CEN429 Lab Ara CA
  Validity
    Not Before: Sep 23 06:01:46 2026 GMT
    Not After : Dec 22 06:01:46 2026 GMT
  Subject: CN=localhost
  Subject Public Key Info:
    Public Key Algorithm: id-ecPublicKey
      Public-Key: (256 bit)
      ASN1 OID: prime256v1
      NIST CURVE: P-256
  X509v3 extensions:
    X509v3 Basic Constraints:
      CA:FALSE
    X509v3 Key Usage: critical
      Digital Signature
    X509v3 Extended Key Usage:
      TLS Web Server Authentication
    X509v3 Subject Alternative Name:
      DNS:localhost, IP Address:127.0.0.1
    X509v3 Subject Key Identifier:
      EA:59:FF:05:1B:BA:D2:ED:D3:B8:2D:CF:12:32:F6:52:27:CD:98:87
    X509v3 Authority Key Identifier:
      F5:BA:39:86:1F:A1:D0:79:C8:DA:83:7D:AE:2C:87:9D:56:45:CE:25
  Signature Algorithm: ecdsa-with-SHA256
  Signature Value:
    30:46:02:21:00:8e:a3:91:48:0a:d5:8a:9d:a7:ee:08:ae:27:...
```

Line by line: **Version 3** shows we can use the extension fields (the X509v3 block). The **Serial Number** is unique among the certificates this CA has issued — when a certificate is revoked, this number is listed in the CRL (Section 10). The **Issuer** (CN=CEN429 Lab Ara CA) tells you **who signed** this certificate — this is where the answer to Section 7's "Question 1" lives. **Validity** is Section 7's "Question 2." The **Subject** (CN=localhost) is the identity the certificate **claims** to belong to — but you must check the SAN before trusting this claim (see above). **Subject Public Key Info** is the public key this certificate **validates** itself (a point on the 256-bit P-256 curve). **Basic Constraints: CA:FALSE** is the answer to Section 7's "Question 3": this certificate cannot sign another certificate. **Subject Alternative Name** is the answer to Section 7's "Question 4." **Authority Key Identifier** is the fingerprint of the public key of the CA that signed this certificate; the client uses it, while building the chain, to avoid the confusion of "a fake intermediate certificate with the same issuer name but signed with a

different key." At the bottom, **Signature Value** is the signature produced over all these fields (the TBSCertificate) **with the CA's private key** — exactly the signing mechanism we learned in Section 5.

Worked example: how many bytes is a certificate?

```
openssl x509 -in sunucu.crt -outform DER | wc -c
wc -c sunucu.crt
```

The DER (binary) form is generally around 500–900 bytes (since curve-based keys are smaller than RSA); the PEM form is roughly 33% larger than this because of Base64 encoding (Base64 turns every 3 binary bytes into 4 text characters). In a TLS handshake, the certificate **chain** the server sends (leaf + intermediate) is the sum of these sizes; this is why a chain using RSA-3072 certificates can be **many times larger** than an ECC (P-256/Ed25519) chain — the reflection, in PKI, of Section 4's size comparison.



Common mistake: trusting the **CN** in **Subject**

The `Subject: CN=localhost` field is only a **claim**; the CA signed the certificate because it verified this claim, but the client must perform **name checking** against the SAN, not this field (see the warning box above). Old code samples and some teaching materials still show code that looks at the CN; this is an approach abandoned with RFC 6125.



Rule

When inspecting a certificate by hand, always use `openssl x509 -noout -text` (or `-ext subjectAltName`) and read the **SAN, validity, basicConstraints, and issuer/subject chain** quartet (Section 7) together; don't look at a single field and decide.

Section 7 explained the **trust a certificate carries**; this section showed **in which bytes** that trust is encoded. The next section switches to the CA's role and **builds this document by hand**.

9. Building a certificate chain with OpenSSL

The steps below build a three-layer chain in your own lab. All files are created in the working folder; **never add the root certificate to any trust store**.

Zincir doğrulamanın dört sorusu

+ ad denetimi (SAN) — beşi birden geçmeden kabul edilmez



Eksik ara sertifika zinciri kurdukmaz; fail-open ile “geçti” saymak en sık hatadır.

1. Root CA (self-signed)

```
openssl genpkey -algorithm EC -pkeyopt ec_paramgen_curve:P-256 -out kok.key
openssl req -x509 -new -key kok.key -sha256 -days 3650 -out kok.crt \
  -subj "/CN=CEN429 Lab Kok CA" \
  -addext "basicConstraints=critical,CA:TRUE" \
  -addext "keyUsage=critical,keyCertSign,cRLSign"
```

2. Intermediate CA (signed by the root, path length 0: no CA can exist beneath it)

```
openssl genpkey -algorithm EC -pkeyopt ec_paramgen_curve:P-256 -out ara.key
openssl req -new -key ara.key -out ara.csr -subj "/CN=CEN429 Lab Ara CA"
openssl x509 -req -in ara.csr -CA kok.crt -CAkey kok.key -CAcreateserial -days 1825 -sha256 \
  -out ara.crt -extfile <(printf
"basicConstraints=critical,CA:TRUE,pathlen:0\nkeyUsage=critical,keyCertSign,cRLSign")
```

3. Server certificate (signed by the intermediate CA, with SAN)

```
openssl genpkey -algorithm EC -pkeyopt ec_paramgen_curve:P-256 -out sunucu.key
openssl req -new -key sunucu.key -out sunucu.csr -subj "/CN=localhost"
openssl x509 -req -in sunucu.csr -CA ara.crt -CAkey ara.key -CAcreateserial -days 90 -sha256 \
  -out sunucu.crt -extfile <(printf
"basicConstraints=CA:FALSE\nkeyUsage=critical,digitalSignature\nextendedKeyUsage=serverAuth\nsubjectAltName=DNS:localhost,IP:127.0.0.1")
```

4. Verify and inspect

```
openssl verify -CAfile kok.crt -untrusted ara.crt sunucu.crt      # sunucu.crt: OK
openssl x509 -in sunucu.crt -noout -text | less                 # inspect the fields
openssl verify -CAfile kok.crt sunucu.crt                       # without the intermediate: ERROR
(chain incomplete)
```

On Windows

The `<( ... )` process-substitution syntax is specific to Bash. On Windows PowerShell, write the extensions to a file (`ara.ext`, `sunucu.ext`) and pass it as `-extfile ara.ext`; the rest of the commands are the same.

The last command producing an error is instructive: a client can only build the chain if the intermediate certificate is also presented. Servers sending only the leaf certificate and forgetting the intermediate is a configuration mistake commonly seen in the field.

Class activity: breaking the chain

Note the `openssl verify` output at every step:

1. Sign the server certificate with the **root** but leave the intermediate CA in the chain. What changes?
2. Produce the intermediate CA with `CA:FALSE` and sign the server certificate with it. What does validation say?
3. Produce a certificate that expires immediately with `-days 0`.
4. Set the SAN to `DNS:baska.ornek` and try it with `openssl s_client -verify_hostname localhost`.

Worked example: solving two scenarios of the class activity from start to finish

Let's solve the first two of the four scenarios above together; you will complete scenarios 3 and 4 yourself.

Scenario 1 — Signing the server certificate directly with the root (skipping the intermediate CA):

```
openssl x509 -req -in sunucu.csr -CA kok.crt -CAkey kok.key -CAcreateserial -days 90 -sha256 \
-out sunucu_kok.crt -extfile sunucu.ext
openssl verify -CAfile kok.crt -untrusted ara.crt sunucu_kok.crt
```

```
sunucu_kok.crt: OK
```

This may be surprising: even though we also supplied `ara.crt`, validation **passed**, because OpenSSL could build the chain from `sunucu_kok.crt` **directly** to `kok.crt`; `ara.crt` was never used on this path, it just sat in the pool as an available candidate. The lesson here is **operational, not cryptographic**: mathematically, the root CA can directly sign any certificate it wants. But we **must not do this**, because if the root CA's private key sits on a system used every day (the "why an intermediate CA" explanation in Section 7), the risk of compromise increases and the whole chain collapses.

Scenario 2 — Producing the intermediate CA with `CA:FALSE` :

```
printf "basicConstraints=critical,CA:FALSE\nkeyUsage=critical,keyCertSign,cRLSign\n" > ara_yanlis.ext
openssl x509 -req -in ara.csr -CA kok.crt -CAkey kok.key -CAcreateserial -days 1825 -sha256 \
-out ara_yanlis.crt -extfile ara_yanlis.ext

openssl x509 -req -in sunucu.csr -CA ara_yanlis.crt -CAkey ara.key -CAcreateserial -days 90 -sha256 \
-out sunucu_yanlis.crt -extfile sunucu.ext

openssl verify -CAfile kok.crt -untrusted ara_yanlis.crt sunucu_yanlis.crt
```

```
CN=CEN429 Lab Ara CA
error 79 at 1 depth lookup: invalid CA certificate
error sunucu_yanlis.crt: verification failed
```

This time validation was **rejected** — because `ara_yanlis.crt`'s `basicConstraints` is `CA:FALSE`, even though the mathematical signature is correct, OpenSSL sees that this certificate has **no authority to sign another certificate** and treats the chain as invalid at "depth 1" (the middle layer). This is proof that the `Basic Constraints` field from Section 8 is an extension that **actually changes behaviour** — not a fancy text field.

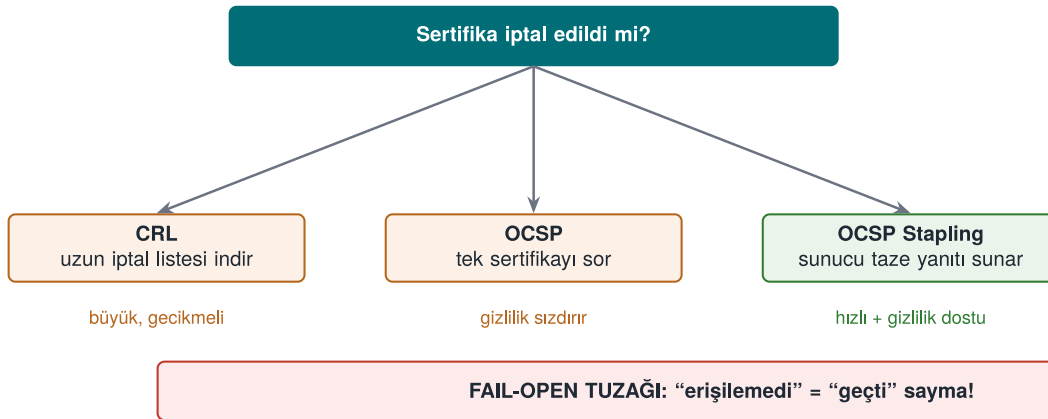
Compare Scenario 1 with Scenario 2: both build a "wrong" chain, but one (**signing directly with the root**) is cryptographically valid and only an operational mistake; the other (**signing with `CA:FALSE`**) is rejected at the protocol level. When an evaluator inspects your PKI setup, this is exactly the distinction they look for: "does the chain technically work" and "does the chain follow the design rules" are two separate questions.

10. Certificate revocation: CRL, OCSP, and stapling (Recipe 10.10–10.12)

If a certificate's private key is stolen, or the certificate was issued incorrectly, it must be **revoked** before its validity period expires. There are three ways for the client to learn this:

Sertifika iptali: CRL · OCSP · zımbalama

“bu sertifika hâlâ geçerli mi?”



Method	How?	Pro	Con
CRL (revocation list)	The CA regularly publishes a signed list of revoked serial numbers	Simple, can be cached offline	The list grows; a revocation is not noticed within the update interval
OCSP	The client asks the CA's responder "is this serial number valid?"	Instant	Latency, privacy (the CA learns which site was visited), what happens if the responder is unreachable?
OCSP stapling	The server attaches a fresh OCSP response for its own certificate to the handshake	Privacy and speed	Requires server configuration

Revocation and CRL check with the lab CA (summary)

```
# producing a CRL with the intermediate CA requires 'openssl ca' configuration; then:
openssl ca -config ara.cnf -revoke sunucu.crt
openssl ca -config ara.cnf -gencrl -out ara.crl
openssl verify -crl_check -CAfile kok.crt -untrusted ara.crt -CRLfile ara.crl sunucu.crt #
"certificate revoked"
```

⚡ Soft-fail

Most browsers **accept** a certificate when they cannot reach the OCSP responder; otherwise a brief outage of the responder would make every site unreachable. But this means an attacker in the middle can bypass revocation checking simply by blocking the OCSP request: the PKI counterpart of [Week 3's fail-open](#) pattern. In critical applications, "a stapled response is mandatory" (OCSP Must-Staple) or short-lived certificates (days, weeks) reduce this problem; this is why certificate lifetimes are getting shorter and shorter.

Worked example: revoking a certificate and validating it against a CRL (end to end)

Let's trace the full version of the "summary" commands above, actually run, step by step. First, the minimal database and configuration file that `openssl ca` needs is set up (this is a heavily scaled-down version of a real CA):

1. Setting up a minimal CA database

```
mkdir -p demoCA/newcerts
touch demoCA/index.txt          # the "ledger" where every issued/revoked certificate is recorded
echo 1000 > demoCA/crlnumber    # the next CRL's version number
echo 1000 > demoCA/serial       # the next certificate's serial number
```

ara.cnf

```
[ ca ]
default_ca = ara_ca

[ ara_ca ]
dir                = demoCA
database           = $dir/index.txt
new_certs_dir      = $dir/newcerts
certificate         = ara.crt
private_key        = ara.key
serial             = $dir/serial
crlnumber          = $dir/crlnumber
default_md         = sha256
default_days       = 90
default_crl_days   = 30
policy             = policy_any
email_in_dn        = no

[ policy_any ]
commonName = supplied
```

2. Revoke, generate CRL, verify (real output)

```
openssl ca -config ara.cnf -revoke sunucu.crt
```

```
Using configuration from ara.cnf
Adding Entry with serial number 35E2F68D...76E7 to DB for /CN=localhost
Revoking Certificate 35E2F68D...76E7.
Database updated
```

```
openssl ca -config ara.cnf -gencrl -out ara.crl
openssl verify -crl_check -CAfile kok.crt -untrusted ara.crt -CRLfile ara.crl sunucu.crt
```

```
CN=localhost
error 23 at 0 depth lookup: certificate revoked
error sunucu.crt: verification failed
```

Step by step, what happened: (1) `-revoke` recorded the certificate's **serial number** (Section 8's `Serial Number` field — here `35E2F68D...76E7`) as "revoked" in the `index.txt` ledger; the certificate itself was not changed or deleted in any way — only a **record** was added. (2) `-gencrl` collected all the revocation records in this ledger and produced a list (`ara.crl`) **signed with the intermediate CA's own private key** — the CRL itself is also a signed document, protected against forgery by the same mechanism from Section 5. (3) With `-crl_check`, validation now checked not only the chain but also the CRL, saw that the serial number was on the list, and rejected it with the "**certificate revoked**" error — even though the certificate's validity date has not yet expired.

This is the complete solution to Exercise 6; repeat the same steps in your own lab and inspect the contents of `index.txt` (`cat demoCA/index.txt`) — every line carries a certificate's record (status, date, serial number, subject name).

Worked example: how big does a CRL get?

If a CA issues 10,000 certificates a year and 1% of them (100) are revoked, since every CRL entry carries ~40–60 bytes (serial number + revocation date + reason code), the CRL size is ~4–6 KB — that looks small. But for a large public CA (millions of certificates, hundreds of thousands of revocations), the CRL can reach **tens of megabytes**; every client downloading this periodically is not practical. OCSP solves this scaling problem by "asking a single question" instead of "downloading the entire list" — but then the privacy and availability trade-off above reappears.



Common mistake: testing revocation checking only in a development environment and taking it to production as-is

A client written without `-crl_check` (or the OCSP equivalent) turned on accepts, without any problem, a certificate that has not expired but **has been revoked**. Because a CRL/OCSP server is usually not set up in a test environment, this mistake is easy to overlook, and it is carried into production "with revocation checking off." Rule: make revocation checking an **explicit configuration option**, and confirm on your deployment checklist that it is turned on.



Rule

The shorter a certificate's lifetime, the less you need a revocation mechanism — this is why the industry (Let's Encrypt and similar) is moving toward shortening certificate lifetimes. If you use long-lived certificates, **make sure** to enable and test CRL or OCSP stapling.

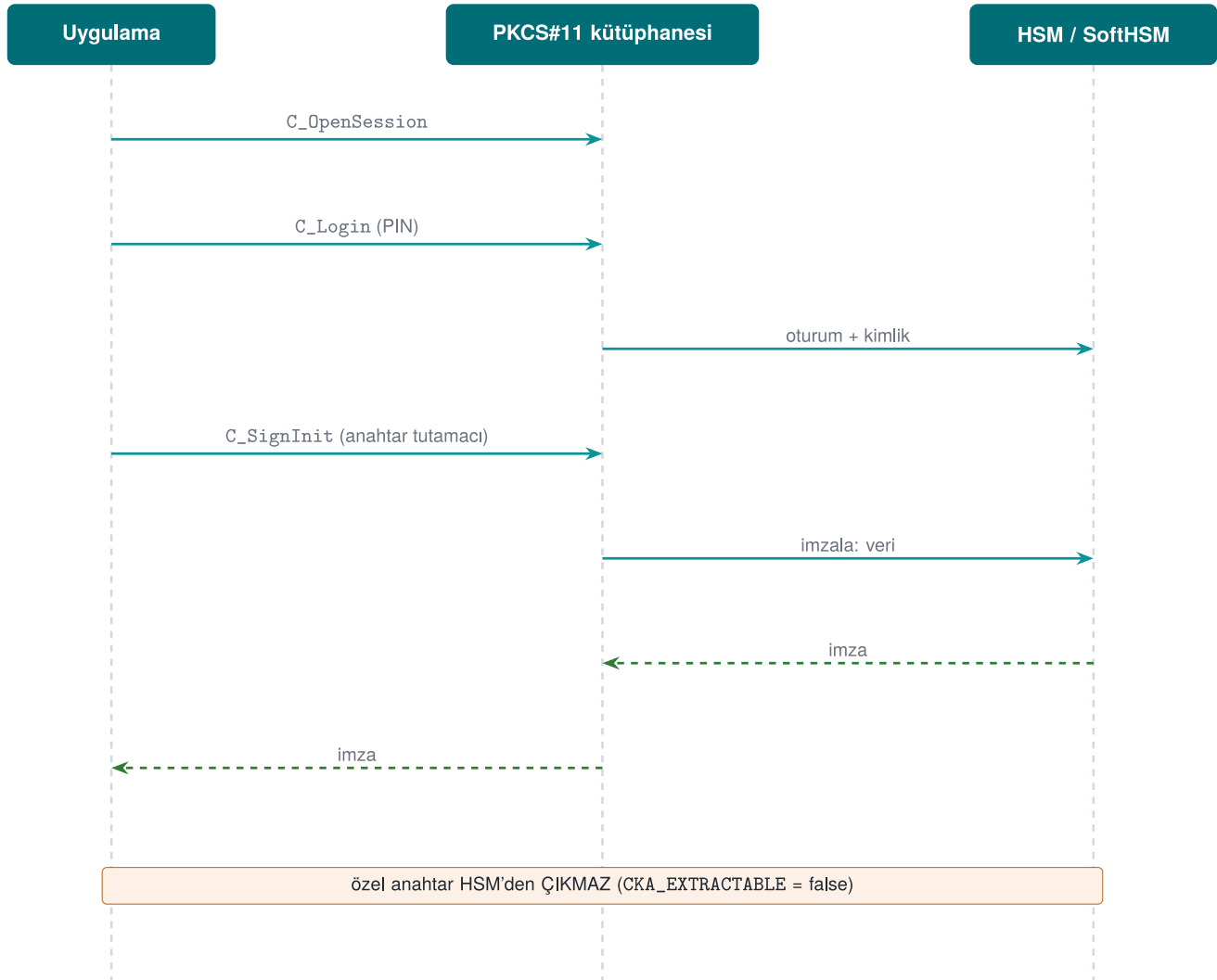
In Section 9 we learned to **build** a chain; in this section we learned to **invalidate** one link in a chain. The next section covers how we protect the most valuable piece of this chain — the private keys themselves.

11. Storing the key in hardware: HSM, PKCS#11, and SoftHSM

A stolen private key destroys all of PKI's guarantees. This is why valuable keys (CA keys, payment keys, code signing keys) are generated inside **hardware security modules** (HSM) and never leave them. The application never sees the key; it tells the HSM "sign this with that key."

PKCS#11 ile imzalama: anahtar hiç dışarı çıkmaz

uygulama anahtarı değil, anahtarın TUTAMACINI görür



Anahtar çalınamaz; ama HSM'e erişimi olan imzalatabilir — erişim denetimi yine şarttır.

Concept	Explanation
HSM	Tamper-resistant, certified (FIPS 140-3, Common Criteria) hardware; does not export the key
PKCS#11 (Cryptoki)	The standard C API for accessing HSMs, smart cards, and cryptographic tokens
Object and attributes	Keys are marked "not exportable" with attributes such as <code>CKA_SENSITIVE</code> , <code>CKA_EXTRACTABLE=false</code>
Session and PIN	The application opens a session to a slot and authenticates with the user's PIN
SoftHSM	An open-source HSM simulation that provides the PKCS#11 interface in software; for development and testing

Generating and listing a key with SoftHSM (WSL/Linux)

```
export SOFTHSM2_CONF=$PWD/softhsm2.conf
mkdir -p belirteciler && echo "directories.tokendir = $PWD/belirteciler" > softhsm2.conf
softhsm2-util --init-token --free --label "cen429" --pin 1234 --so-pin 5678
pkcs11-tool --module /usr/lib/softhsm/libsofthsm2.so --login --pin 1234 \
    --keypairgen --key-type EC:prime256v1 --label imza-anahtari
pkcs11-tool --module /usr/lib/softhsm/libsofthsm2.so --login --pin 1234 --list-objects
```

⚠ SoftHSM is not an HSM

SoftHSM stores keys on disk, in a file; there is no hardware protection. Its value is that it lets you write your application **against the PKCS#11 interface**: the same code connects, in production, to a real HSM or smart card just by changing the module path. We will look at software-based security modules again from this angle in [Week 11](#).

🔧 How it's done in the field

In payment systems, the keys at the top of the key hierarchy (the card issuer's master keys, the keys that generate device keys) sit in HSMs on the server side; in the mobile payment platform's back end, the HSM appears as a separate component in the architecture diagram ([the example architecture from Week 1](#)). The keys that land on the phone are generated on the server with an HSM, encrypted for binding to the device, and the library itself never **generates** an account key. The guide contains an algorithm inventory that lists every algorithm it uses together with the standard it rests on (NIST FIPS, ISO/IEC 9797-1, EMV specifications).

End to end: how is a signing request processed inside an HSM?

Let's make the sentence "the key never leaves" concrete. The [PKCS#11](#) standard (OASIS Cryptoki v3) defines the following steps for an application to have an HSM sign something **without ever sending or receiving a raw key byte**:

1. **Open a session:** The application connects to a "slot" in the HSM with `C_OpenSession`.
2. **Authenticate:** The PIN is sent with `C_Login`; if the PIN is **wrong**, the HSM locks itself after a certain number of attempts (hardware-level protection against brute force — an easy safeguard to forget in software).
3. **Find the key:** The **handle** of a previously generated key object (a reference number inside the HSM, not the key itself) is obtained with `C_FindObjectsInit` / `C_FindObjects`.
4. **Sign:** The call `C_SignInit + C_Sign(handle, data, ...)` sends the data to the HSM; the HSM processes the data with the key **inside itself** and returns only the **signature**. The private key's bytes **never** enter the application code's memory at any point.
5. **Close the session:** `C_Logout` / `C_CloseSession`.

The shape of the PKCS#11 C_Sign call (OASIS Cryptoki v3, reference)

```
CK_RV C_Sign(
    CK_SESSION_HANDLE hSession, /* open session */
    CK_BYTE_PTR       pData,    /* data to be signed */
    CK_ULONG           ulDataLen,
    CK_BYTE_PTR        pSignature, /* output buffer filled by the HSM: only the signature, not the key
*/
    CK_ULONG_PTR       pulSignatureLen
);
```

Compare this flow with an ordinary OpenSSL call (`EVP_DigestSign` , Section 5): both **interfaces** carry the same idea ("give the data, get the signature"), but in OpenSSL the key (if read from a file) **sits in process memory**; in **PKCS#11** the key **never** enters the application's address space. Moving Section 5's `imza_dogrula` function to an HSM only means routing the signing side to a `C_Sign` call – the **verification** side already only needs the public key, so it does not need the HSM.

Why is `CKA_EXTRACTABLE` the most critical attribute?

When a **PKCS#11** key object is generated (`C_GenerateKeyPair`), the attributes `CKA_EXTRACTABLE=false` and `CKA_SENSITIVE=true` are given **at key-generation time** and **cannot be changed afterward**. These two flags prevent, **at the hardware level**, the HSM from ever giving out the key in plaintext, even if a function like `C_WrapKey` is called.

⚡ Common mistake: generating a key with default attributes

Some **PKCS#11** libraries and tools leave `CKA_EXTRACTABLE` as **true** unless explicitly specified (for compatibility). In this case, even though the key is technically generated in an HSM, someone with the right permissions can **export** it – the assurance "we use an HSM" becomes an empty sentence. In code review, look for `CKA_EXTRACTABLE = CK_FALSE` being given **explicitly** in the attribute template of the `C_GenerateKeyPair` call.

✓ Rule

When generating a valuable key (a CA root key, a code-signing key, a payment master key), `CKA_EXTRACTABLE=false` and `CKA_SENSITIVE=true` must be explicitly set, and right after generation you must **verify** with `pkcs11-tool --list-objects` (or its equivalent) that these attributes were actually applied – don't trust the default.

We will look again, in **Week 11**, at software-based security modules (structures like the Secure Enclave/TEE on mobile devices) through the **PKCS#11** logic of this section; the principle you learned here – "work through a handle, never move the raw key" – is the foundation of that entire discussion. The next and final technical section looks at why the RSA/ECC we consider secure today needs to change in the future.

12. A look at post-quantum cryptography

A sufficiently large quantum computer can efficiently solve the mathematical problems RSA and elliptic curves rest on (factoring, discrete logarithm). Such a computer does not exist yet; but the idea that encrypted traffic recorded today could be decrypted years from now ("harvest now, decrypt later") puts the transition on the agenda starting today.

Kuantum sonrası: neyi değiştiriyor, neyi değiştirmiyor?

NIST seçimi: ML-KEM (Kyber) ve ML-DSA (Dilithium)

TEHDİT ALTINDA
RSA, DH, ECDH, ECDSA
"şimdi topla, sonra çöz"
bugünkü trafik ileride çözülebilir
geçiş bugün planlanır

BÜYÜK ÖLÇÜDE GÜVENLİ
AES-256 (anahtar boyu yeter)
SHA-384 / SHA-3
simetrik dünya ayakta
yalnız anahtar boyu artar

Uzun ömürlü sırlarınız varsa hibrit anahtar değişimine geçiş bugünün kararıdır.

Item	What's changing?
Symmetric ciphers and digests	Limited impact: AES-256 and SHA-384/512 are considered sufficient
Key agreement	NIST standardized ML-KEM (FIPS 203) in 2024; hybrid use alongside X25519 is becoming widespread in TLS
Signature	The ML-DSA (FIPS 204) and SLH-DSA (FIPS 205) standards have been published
Timeline	NIST's transition plan foresees phasing out RSA and elliptic-curve algorithms after 2030

For this course's project, the practical outcome is a single sentence: **crypto agility**. Don't hard-code algorithm names everywhere in your code; use formats that carry a key version and algorithm identifier (the key version field from [Week 3](#)), so that when the algorithm changes, only configuration and re-encryption are needed.

Worked example: generating post-quantum keys today

OpenSSL 3.5 supports the standardized post-quantum algorithms **directly**; you can generate them today, on your own machine, and compare their sizes with classical algorithms:

ML-KEM-768 (key agreement) and ML-DSA-65 (signature) — real output

```
openssl genpkey -algorithm ML-KEM-768 -out mlkem.pem
openssl pkey -in mlkem.pem -pubout -out mlkem_pub.pem
wc -c mlkem_pub.pem

openssl genpkey -algorithm ML-DSA-65 -out mldsa.pem
openssl pkey -in mldsa.pem -pubout -out mldsa_pub.pem
echo hello > belge.txt
openssl pkeyutl -sign -inkey mldsa.pem -rawin -in belge.txt -out belge.mldsa.sig
wc -c mldsa_pub.pem belge.mldsa.sig
openssl pkeyutl -verify -pubin -inkey mldsa_pub.pem -rawin -in belge.txt -sigfile belge.mldsa.sig
```

```
1714 mlkem_pub.pem
2770 mldsa_pub.pem
3309 belge.mldsa.sig
Signature Verified Successfully
```

Compare with the table in Section 4:

Algorithm	Public key (PEM, bytes)	Signature (bytes)	Security basis
Ed25519 (classical)	116	64	Elliptic-curve discrete logarithm
RSA-3072 (classical)	636	384	Factoring
ML-DSA-65 (PQC)	2,770	3,309	Lattice-based problem (Module-LWE)
ML-KEM-768 (PQC, key agreement)	1,714	—	Lattice-based problem (Module-LWE)

The difference is striking: ML-DSA-65's public key is **~24 times** larger than Ed25519's, and its signature is **~52 times** larger. This shows that the move to PQC is not just "changing an algorithm name" — many places, such as the TLS handshake, certificate size, and the number of network packets, must account for this growth. NIST's recommended **hybrid** approach for TLS (like `X25519MLKEM768`, the version of Section 6's X25519 agreement run together with PQC) exists precisely to manage this transition cost: classical and PQC keys are used **at the same time**; if the classical side is broken, PQC still protects; if the PQC side (new, not-yet-proven math) is broken, the classical side keeps protecting.

Why lattice problems? A short intuition

RSA's security rests on "the difficulty of factoring a large number," ECC's security rests on "the difficulty of taking a discrete logarithm on a curve" — Shor's algorithm efficiently solves **both** on a sufficiently large quantum computer. ML-KEM and ML-DSA instead rest on **lattice** problems: in a multi-dimensional space, in a structure formed by regularly spaced points (a lattice), "finding the lattice point closest to a given point" or "solving a system of equations with small error terms" (Learning With Errors, LWE) is computationally very hard at high dimensions, and **so far**, no efficient solution has been found for either classical or quantum computers. "Not having been found so far" does not mean "will never be found" — which is why the hybrid approach is recommended.



Common mistake: saying 'there's no quantum computer yet, no rush'

Traffic encrypted and recorded today with RSA/ECC (e.g., an intelligence record, a long-term medical record, a government document that must stay confidential for 20–30 years) can be recorded today and decrypted once a quantum computer matures, under the "harvest now, decrypt later" strategy. For a system with a confidentiality window of 10–20 years, the transition decision must be made **today**, not once a quantum computer exists.



Rule

If you are designing a new protocol, support a **hybrid** (classical + PQC) key agreement from now on where possible, or at least, with the crypto-agility principle from Section 1 (formats carrying an algorithm identifier/version), make a future transition to PQC possible **without a rewrite**.

This section was the **final stop** on the map we drew in Section 1: which algorithm, which problem, how secure, until when. From Section 13 onward you pour this knowledge into your own project's algorithm inventory.

13. Term project: this week

Draft the **S8** and **S11** sections expected at the final checkpoint:

- ✓ **S8 — Algorithm inventory:** for every algorithm used in your project, its purpose, key length, mode, the standard it rests on, and the library. If the table contains an obsolete algorithm, give its justification and a migration plan.
- ✓ **S8 — Key lifecycle table:** for every key, where it is generated, where it is stored, its crypto-period, and its renewal and destruction ([Week 3](#)).
- ✓ **S11 — Secure communication:** TLS version and validation; if pinning is used, the SPKI and backup pin; message-level protection.
- ✓ If your project has an update or configuration file, add **signature verification** (Ed25519 recommended) and reject rollbacks.
- ✓ Optional: use this week's root → intermediate → leaf chain for your project's test certificates; **do not** put the root key in the repository.

Worked example: filling in the S8 table for a sample component

Facing a blank table, it can be unclear what to write; let's see how S8 would be filled in for a sample project component ("user session token generation") — replace the rows with your own components in your own project:

Job	Algorithm	Key length	Mode/Padding	Library	Standard it rests on	Note
Session data encryption	AES-256	256 bit	GCM (AEAD)	OpenSSL EVP 3.x	NIST SP 800-38D	Section 2
Session integrity	— (GCM tag)	—	—	OpenSSL EVP	SP 800-38D	Inside AEAD, no separate MAC
Key agreement	X25519	128-bit level	ECDH	OpenSSL EVP	RFC 7748	Section 6
Server authentication	ECDSA P-256	128-bit level	—	OpenSSL/TLS	FIPS 186-5	TLS certificate, Sections 7–9
Update package signature	Ed25519	128-bit level	—	OpenSSL EVP	RFC 8032	Section 5, version field signed

The key lifecycle table (this week's counterpart to [Week 3's key management topic](#)) for the same component:

Key	Where generated	Where stored	Crypto-period	Renewal	Destruction
Session key (AES-256)	Server, on every session	In memory only	Single session	New on every session	Erased from memory when the session closes
TLS server private key	Server, HSM/SoftHSM	HSM (Section 11)	90 days (certificate lifetime)	Automatic renewal (ACME)	Never leaves the HSM, via <code>CKA_EXTRACTABLE=f</code> <code>alse</code>
Update signing key	Offline, via a ceremony	HSM, offline backup	For the life of the product	Annual review	Per organizational policy

These two tables bring together, in a single place, every concept you've seen in Sections 1–12 of this week ("which mode," "which standard," "where it's stored," "when it's renewed") – an evaluator, at the final checkpoint, looks for exactly these two tables.

When adapting the table to your own project

Tie every row to a **real** call in your own codebase (you can even give the file name and line number); write a traceable reference like "`auth.c:142, EVP_EncryptInit_ex` with AES-256-GCM" instead of a generic sentence like "we use AES." This is the final-checkpoint counterpart of Section 1's rule that "the algorithm name alone is not enough."

14. Work on your own

Exercise 1 – Easy: HMAC and length extension

For the same message and key, compute an HMAC with `openssl dgst -sha256 -mac HMAC` and a plain digest with `printf 'anahtar+mesaj' | openssl dgst -sha256`. Explain why the plain $H(K \parallel m)$ construction is insecure, using the concept of length extension.

Exercise 2 – Easy: comparing RSA and Ed25519

Generate a 3072-bit RSA key and an Ed25519 key. Compare the sizes of the public key files, the signatures, and the time it takes to perform 100 signing operations (`openssl speed rsa3072 ed25519` can also be used).

Exercise 3 – Medium: the return value of signature verification

Run the `imza_dogrula` function from the "Digital signature" section. Try it first with a correct signature, then with a signature with one byte changed, then with an invalid key object. Show under which condition the difference between `if (sonuc)` and `if (sonuc == 1)` turns into a security vulnerability.

? Exercise 4 – Medium: breaking the chain

Carry out the four scenarios from the "Class activity" section and record the `openssl verify` and `openssl s_client` error messages for each one in a table.

? Exercise 5 – Medium: TLS server and the chain

Start a local TLS server with the server certificate you produced (`openssl s_server -cert sunucu.crt -key sunucu.key -cert_chain ara.crt -accept 8443`) and connect with `openssl s_client -connect localhost:8443 -CAfile kok.crt -verify_return_error` . What happens if the server does not send the intermediate certificate?

? Exercise 6 – Medium: revocation

Prepare an `openssl ca` configuration for your lab's intermediate CA, revoke the server certificate, produce a CRL, and show, with `openssl verify -crl_check` , that the revocation is detected.

? Exercise 7 – Hard: a replay-resistant message

Write two programs: one encrypts a message in the form `counter || data` with AES-GCM (let the counter be the AAD) and sends it, the other receives it and rejects a counter it has seen before or one that is smaller. Resend a recorded message and show that it is rejected.

? Exercise 8 – Hard: signing with PKCS#11

Sign a file with `pkcs11-tool --sign` using the key you generated in SoftHSM, and verify the signature with OpenSSL from the public key. Try to export the private key from SoftHSM; how does the `CKA_EXTRACTABLE` attribute prevent this?

15. Self-check

? 1. What is the key length for RSA and for elliptic curve at the 128-bit security level?

RSA 3072 bits, elliptic curve 256 bits (P-256, Ed25519/X25519).

? 2. Under what condition is the padding oracle attack possible? Give two defences.

When the decoder reports "invalid padding" in a way distinguishable from other errors (a different message or timing). Using AEAD; if CBC is mandatory, encrypt-then-MAC and verifying the MAC before the padding, and returning all errors in the same form.

? 3. Why is HMAC used instead of $H(K \parallel m)$?

In Merkle–Damgård-structured digests, $H(K \parallel m)$ is exposed to length-extension attacks; HMAC's nested construction prevents this.

? 4. Why is encrypt-then-MAC the correct order?

The MAC is computed over the ciphertext and verified before decryption; an invalid message is never decrypted, so no padding oracle or plaintext leak occurs.

? 5. Why doesn't a MAC alone prevent replay? How is it prevented?

A MAC proves the message was not altered, not that it is new. A counter, timestamp, or a server-issued one-time value is included in the MAC, and the receiver rejects repeats.

? 6. Which padding schemes are recommended for RSA encryption and signing?

OAEP (SHA-256) for encryption, PSS for signing.

? 7. Why is reusing the same random value in two ECDSA signatures a disaster?

The private key can be computed from the two signatures. Ed25519 or deterministic ECDSA (RFC 6979) removes this risk.

? 8. Why is checking the result of `EVP_DigestVerify` with `if (sonuc)` incorrect?

Negative error values are also counted as true; on an error the signature is treated as valid. The check must be `== 1`.

? 9. Why is unauthenticated Diffie–Hellman open to man-in-the-middle? How does TLS 1.3 fix it?

DH does not say who you agreed with; someone in the middle can perform DH separately with each party. In TLS 1.3, the server signs the handshake digest — which also includes the ephemeral DH values — with the key in its certificate; the client validates the certificate.

? 10. Why is the root CA kept offline, and why does an intermediate CA exist?

Compromise of the root key collapses the entire chain, and removing the root from trust stores takes years. The intermediate CA does the day-to-day work; if it is compromised, only that one is revoked.

? 11. Why is name checking done against the SAN, not the CN?

Current standards and clients consider only the SAN extension; looking at the CN is an old and ambiguous method.

? 12. What is 'soft-fail' in OCSP, and why is it dangerous?

Accepting the certificate when the responder cannot be reached. An attacker in the middle can bypass revocation checking by blocking the OCSP request (fail-open). Mandatory stapling or short-lived certificates reduce this.

? 13. How does SoftHSM differ from a real HSM? Why is it used anyway?

It stores keys on disk, with no hardware protection. It lets you write your application against the [PKCS#11](#) interface; in production, the same code connects to a real HSM.

? 14. What is crypto agility, and how does it relate to the post-quantum transition?

Keeping algorithms replaceable through formats that carry a version and algorithm identifier, instead of hard-coding them everywhere in the code. Moving to new algorithms like ML-KEM and ML-DSA can then be done with only configuration and re-encryption.

? 15. Explain why brute-forcing an AES key at the 128-bit security level is impractical, even with a hypothetical computer capable of 10^{18} operations per second.

2^{128} operations are required ($\approx 3.40 \times 10^{38}$). Even at this speed it takes $\approx 1.08 \times 10^{13}$ years — about 780 times the age of the universe ($\approx 1.38 \times 10^{10}$ years). This is why AES-128 is considered secure against brute force today and for the foreseeable future.

? 16. If a 13-byte message is encrypted in a 16-byte block under [PKCS#7](#) padding, how many bytes is the padding and what is its value?

3 bytes; each with the value `03` ($16 - 13 = 3$, as many bytes as are missing, each with that value).

? 17. Why do repeating plaintext blocks in ECB mode produce repeating ciphertext blocks? How does CBC prevent this?

ECB encrypts every block independently with the same key; the same input always gives the same output. CBC chains each block by XORing it with the previous one (the IV for the first block); so identical plaintext blocks produce different ciphertext depending on the different preceding blocks.

? 18. How much of the tag does a single-character change alter in HMAC? What is this property called? ✓

Roughly half. This is called the avalanche effect.

? 19. Why is a constant-time function used instead of `memcmp` when comparing a MAC/signature? ✓

`memcmp` stops at the first differing byte; this makes the comparison time proportional to the number of correct bytes. An attacker who can make enough attempts can discover the correct value byte by byte from the difference in response time (a timing attack). A constant-time function (like `CRYPTO_memcmp`) always walks through all the bytes.

? 20. Why do you get different ciphertexts when you encrypt the same plaintext twice with RSA-OAEP? ✓

OAEP mixes a random value into the padding on every encryption. Raw/unpadded RSA is deterministic ($c = m^e \bmod n$) and always gives the same output for the same input; this is why raw RSA is never used directly for encryption.

? 21. How do you verify that Alice and Bob reach the same shared secret with X25519? Should this secret be used directly as an AES key? ✓

`pkeyutl -derive` is run separately on both sides and the output files are compared (`cmp`); both reach the same secret. No — raw DH output is not uniformly random and is generally a single value; it must first be passed through a KDF (HKDF) to derive as many keys as needed.

? 22. What happens when `openssl verify` is given both a correct and an irrelevant intermediate certificate? ✓

OpenSSL builds the chain along any valid path it can (e.g., directly to the root); the irrelevant certificate just remains an unused candidate, and validation can still succeed. This is an operational, not a cryptographic, situation.

? 23. What happens if an intermediate CA certificate's `basicConstraints` field is `CA:FALSE`? ✓

This certificate no longer has the authority to sign other certificates; validation of any certificate signed with it is rejected with an "invalid CA certificate" error.

? 24. Does `openssl ca -revoke` delete a certificate from disk? ✓

No. The certificate file does not change; only a revocation record is added to the CA's `index.txt` ledger. This record is later turned into a signed CRL with `-gencrl`.

? 25. What does the `CKA_EXTRACTABLE=false` attribute do, and when must it be set? ✓

It prevents an [HSM/PKCS#11](#) key from being exported in plaintext, at the hardware level. It must be set explicitly when the key is generated (`C_GenerateKeyPair`); it cannot be changed afterward.

? 26. State how many times larger the ML-DSA-65 signature is than the Ed25519 signature (in this week's measurement), and explain the practical consequence. ✓

About 52 times larger (3,309 bytes / 64 bytes). The TLS handshake, certificate size, and network traffic all grow; this is why hybrid approaches and crypto agility matter during the transition.

16. Resources and further reading

Course textbook — J. Viega, M. Messier, *Secure Programming Cookbook for C and C++*, O'Reilly, 2003:

- Recipe 5.2–5.4 (algorithm, key length, and mode selection), 5.11 (padding), 5.16 (an error-resistant API)
- Recipe 6.3–6.4 (digest and MAC selection), 6.10 (HMAC), 6.18 (combining encryption and integrity), 6.19 (rolling your own MAC — a warning), 6.21 (a MAC against replay)
- Recipe 7.1–7.3 (public-key usage and selection), 7.10–7.14 (RSA encryption, signing, and secure combination), 7.16–7.17 (DER and PEM)
- Recipe 8.15–8.19 (password-based key exchange, authenticated key exchange with RSA, Diffie–Hellman, DH + signature, authentication without PKI)
- Recipe 10.1, 10.4–10.5 (PKI and X.509 validation), 10.10–10.12 (CRL and OCSP)

Open standards and resources

- NIST SP 800-57 Part 1 (key management and security levels), SP 800-131A (algorithm transitions), SP 800-38A/D (modes, GCM), FIPS 186-5 (digital signature), FIPS 203/204/205 (post-quantum).
- RFC 2104 (HMAC), RFC 8017 ([PKCS#1](#) v2.2: OAEP, PSS), RFC 8032 (Ed25519), RFC 7748 (X25519), RFC 5280 (X.509 and CRL), RFC 6960 (OCSP), RFC 6979 (deterministic ECDSA), RFC 8446 (TLS 1.3).
- OASIS [PKCS#11](#) v3; SoftHSM documentation; OpenSSL 3 man pages (`genpkey` , `req` , `x509` , `verify` , `ca` , `pkeyutl`).
- MITRE CWE: CWE-327 (broken algorithm), CWE-326 (insufficient key length), CWE-295 (improper certificate validation), CWE-299 (missing revocation check), CWE-347 (improper verification of a signature), CWE-294 (authentication bypass by replay).



Glossary



Term	Turkish	Short definition
Mode of operation	Çalışma kipi	How a block cipher is applied to long messages
Padding oracle	Dolgu kâhini	Decrypting by distinguishing the padding error
Encrypt-then-MAC	Şifrele-sonra-MAC	Encrypting first, then taking the MAC of the ciphertext
Replay	Yeniden oynatma	Resending a valid old message
OAEP / PSS	—	RSA's secure encryption and signature paddings
Key agreement	Anahtar anlaşması	Two parties producing a shared secret (DH, ECDH)
CA / RA	Sertifika / kayıt otoritesi	The party that issues certificates / verifies identity
SAN	Konu alternatif adı	The domain names and IPs a certificate is valid for
CRL / OCSP	İptal listesi / çevrimiçi durum protokolü	Ways of announcing certificate revocation
Stapling	Zımbalama	The server attaching its OCSP response to the handshake
HSM	Donanım güvenlik modülü	Tamper-resistant hardware that never gives out the key
PKCS#11	—	The standard API for accessing cryptographic tokens
Crypto agility	Kripto çevikliği	Being able to swap out algorithms easily



Next week

Week 11 — Whitebox cryptography. This week we saw the "entrust it to hardware" path for protecting a key (HSM, PKCS#11, `CKA_EXTRACTABLE=false`, §11). Week 11 asks the same question with no hardware available: how do you protect a key inside an application that is pure software, with no access to any HSM? Table-based whitebox AES and the attacks built against it (BGE, DCA, DFA) are the software-side counterpart of this week's "the key must never be exposed" principle.